

Upsell, Cross-Sell & Recommendation Engine

Module 2 — Functional Specification

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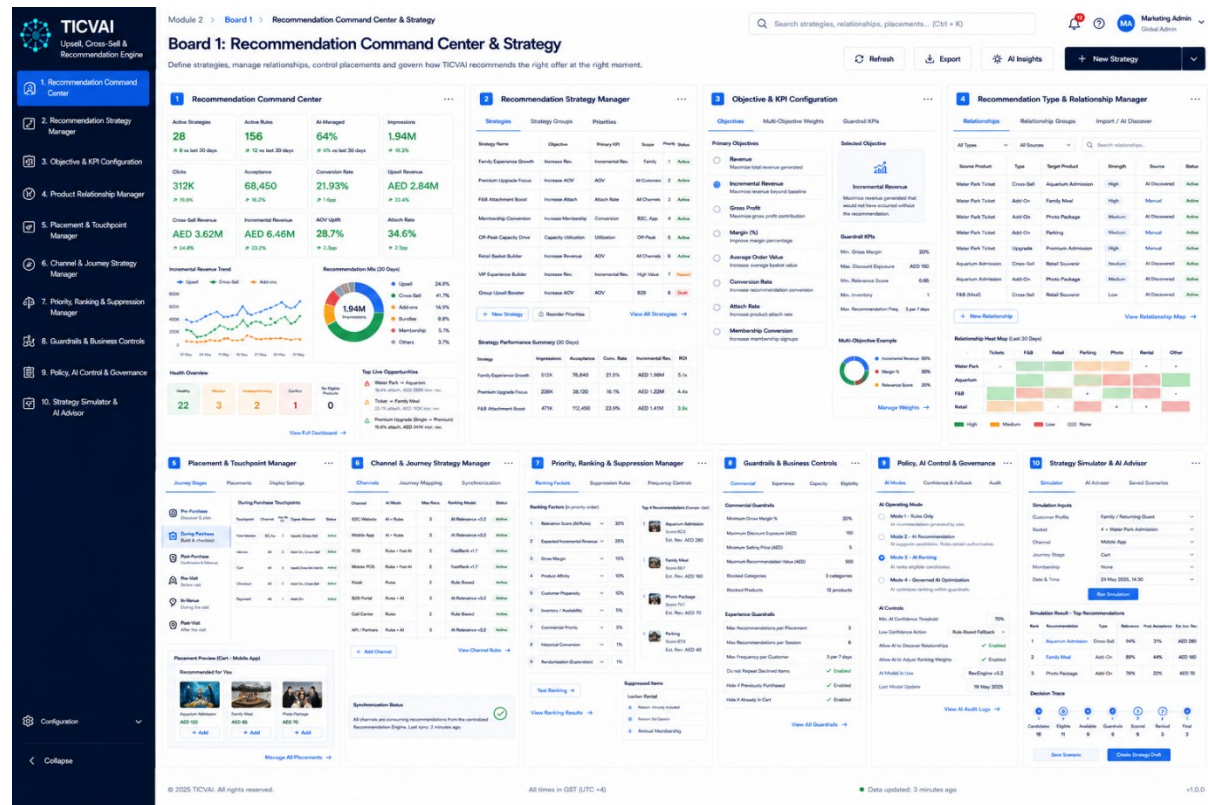
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Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 1 — Recommendation Command Center & Strategy



Board Objective

Board 1 shall establish the **central command, configuration, strategy, monitoring, and governance layer** for TICVAI's Upsell, Cross-Sell & Recommendation Engine.

This board answers the fundamental questions:

What are we trying to recommend? Where should recommendations appear? What commercial objective are we optimizing? Which recommendation strategy should be used? How is AI allowed to participate? And how is the recommendation engine performing?

This is deliberately separated from the Promotions & Bundles module.

The Promotions module determines:

What commercial offers, discounts and bundles exist?

The Recommendation Engine determines:

Which product, upgrade, bundle, add-on or offer should TICVAI present to this particular guest, at this particular moment, through this particular channel?

The underlying matrix requires dynamic upsell/cross-sell throughout the booking journey, post-purchase recommendations, in-venue/location-aware recommendations, membership upgrades, personalized recommendations using CRM and behavioral information, AI-driven recommendations, and synchronized recommendation behavior across channels.

Board 1 establishes the foundation for all of those capabilities.

Board 1 shall contain 10 backend screens.

Screen 1 — Recommendation Command Center

Purpose

Provide the central operational dashboard for the complete TICVAI Recommendation Engine.

This becomes the entry point for Marketing, Commercial, Revenue, CRM and authorized operational users.

KPI Cards

Display:

- Active Recommendation Strategies
- Active Recommendation Rules
- AI-Managed Recommendations
- Recommendation Impressions
- Recommendation Clicks
- Recommendation Acceptance
- Recommendation Conversion Rate
- Upsell Revenue

- Cross-Sell Revenue
- Incremental Revenue
- Average Basket Uplift
- Recommendation Attach Rate

Recommendation Health

Show:

- Healthy
- Monitor
- Underperforming
- Conflict
- Inventory Issue
- No Eligible Product
- AI Warning
- Suspended

Recommendation Mix

Break down by:

- Upsell
- Cross-sell
- Upgrade
- Add-on
- Bundle
- Membership
- Experience
- F&B
- Retail
- Parking

- Photo
- Rental
- External product

Live Recommendation Activity

Example:

Water Park → Aquarium

12,840 recommendations

18.4% acceptance

AED 286K incremental revenue

AI Executive Insight

Example:

Guests purchasing Water Park family tickets have a 34% higher probability of purchasing a Family Meal when the recommendation is displayed immediately after ticket selection rather than during payment.

Screen 2 — Recommendation Strategy Manager

Purpose

Define the commercial strategy TICVAI should use when generating recommendations.

Instead of hard-coding recommendation behavior into B2C or POS, administrators configure reusable strategies centrally.

Strategy Examples

- Maximize Revenue
- Maximize AOV
- Maximize Margin
- Increase Attach Rate
- Increase Experience Participation

- Promote Membership
- Increase F&B Spend
- Increase Retail Spend
- Promote Underutilized Capacity
- Improve Guest Experience

Strategy Configuration

For each strategy:

- Strategy name
- Description
- Business objective
- Business entity
- Venue
- Channel
- Customer segment
- Effective dates
- Recommendation types
- Priority
- Status

Example

Strategy

Family Experience Growth

Objective

Increase cross-attraction sales.

Primary KPI

Incremental Revenue

Secondary KPI

Attach Rate

Target

Families

Strategy Priority

If multiple strategies qualify:

1. Customer Experience
2. Revenue
3. Capacity Optimization
4. Membership Growth
5. F&B Attachment

The order must be configurable.

Screen 3 — Recommendation Objective & KPI Configuration

Purpose

Define exactly what the recommendation engine should optimize.

This is important because the "best" recommendation depends on the business objective.

Optimization Objectives

Support:

- Revenue
- Incremental revenue
- Gross profit
- Margin
- AOV
- Conversion
- Attach rate

- Units per transaction
- Membership conversion
- Upgrade conversion
- Capacity utilization
- Inventory movement

Multi-Objective Strategy

Allow weighted objectives.

Example:

Incremental Revenue — 50%

Margin — 30%

Guest Relevance — 20%

Guardrail KPIs

Administrators may configure:

- Minimum margin
- Maximum discount
- Minimum relevance score
- Minimum inventory
- Maximum recommendation frequency

Example

TICVAI should not recommend a product simply because it produces high revenue if:

Inventory is nearly exhausted.

or:

The recommendation has very low relevance for the customer.

Screen 4 — Recommendation Type & Product Relationship Manager

Purpose

Define the commercial relationships from which recommendations can be generated.

Recommendation Types

Configure:

Upsell

Standard Ticket → Premium Ticket

Upgrade

Single Attraction → Multi-Attraction Pass

Cross-Sell

Ticket → F&B

Add-On

Ticket → Photo Package

Bundle

Ticket → Family Experience Bundle

Membership

Day Ticket → Annual Membership

Ancillary

Ticket → Parking

Relationship Structure

Source Product

↓

Recommendation Relationship

↓

Target Product

Example:

Water Park Ticket

↓

Cross-Sell

↓

Family Meal

Relationship Scope

Support relationships between:

- Ticket → Ticket
- Ticket → Bundle
- Ticket → F&B
- Ticket → Retail
- Ticket → Parking
- Ticket → Locker
- Ticket → Photo
- Ticket → Rental
- Ticket → Membership
- Ticket → Experience
- F&B → Retail
- Retail → F&B
- Membership → Experience
- Product → External Product

Relationship Source

Identify whether the relationship was:

- Manually configured

- Imported
 - Rule generated
 - AI discovered
-

Screen 5 — Recommendation Placement & Touchpoint Manager

Purpose

Define **where in the customer journey recommendations are permitted to appear**.

This is a major requirement for the engine.

Journey Touchpoints

Support:

Pre-Purchase

- Homepage
- Search
- Product page
- Availability selection

During Purchase

- Ticket selection
- Add-ons
- Cart
- Checkout
- Payment preparation

Post-Purchase

- Confirmation page
- Confirmation email
- Mobile App

- Digital ticket
- Pre-visit communication

In-Venue

- Mobile App
- POS
- Kiosk
- Location-aware experience
- Post-redemption journey

Post-Visit

- Follow-up
- Membership conversion
- Return-visit offer

Placement Configuration

For each placement:

- Channel
- Journey stage
- Placement name
- Recommendation types allowed
- Maximum recommendations
- Ranking strategy
- Display format
- Refresh behavior

Example

Cart

Maximum recommendations:

3

Allowed:

Cross-Sell + Upgrade + Add-On

Priority:

1. Upgrade
 2. Experience
 3. F&B
-

Screen 6 — Channel & Journey Strategy Manager

Purpose

Control recommendation behavior across TICVAI's omnichannel environment.

Supported Channels

- B2C Website
- Mobile App
- POS
- Mobile POS
- Kiosk
- B2B
- Call Center
- API
- Reseller
- OTA
- Partner

Channel Configuration

Example:

B2C

Full AI personalization.

Maximum 3 recommendations.

POS

Fast recommendations.

Maximum 2.

No complex customer interaction required.

Kiosk

Visual/simple recommendations.

Maximum 2.

B2B

Contractual and group-related recommendations.

Channel Synchronization

The central engine should ensure:

The same commercial recommendation policy is used across channels.

But presentation can differ by channel.

Example

Recommendation:

Upgrade to Premium Admission

B2C:

Rich card.

POS:

Cashier prompt.

Kiosk:

Upgrade button.

API:

Recommendation object.

Screen 7 — Recommendation Priority, Ranking & Suppression Manager

Purpose

Define how multiple eligible recommendations are ranked before being presented.

This is different from Promotion Board 8.

Promotion Board 8 determines:

Which discounts can coexist.

Recommendation Board 1 determines:

Which products/offers deserve the limited recommendation positions.

Ranking Factors

Support:

- AI relevance
- Revenue potential
- Margin
- Product affinity
- Customer propensity
- Inventory
- Capacity
- Commercial priority
- Campaign priority
- Historical conversion

Example Ranking

Available recommendations:

Recommendation	Relevance	Revenue	Priority
Aquarium	92	High	1

Recommendation	Relevance	Revenue	Priority
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Family Meal	87	Medium	2
Photo Package	74	Low	3
Parking	68	Low	4

If placement supports only 3:

Parking is suppressed.

Suppression Rules

Prevent recommendations when:

- Already purchased
- Already included
- Customer declined repeatedly
- Product unavailable
- Capacity unavailable
- Product conflicts
- Customer not eligible
- Recommendation frequency exceeded

Screen 8 — Recommendation Guardrails & Business Controls

Purpose

Protect the business and customer experience from inappropriate AI or rule-generated recommendations.

Commercial Guardrails

- Minimum selling price
- Minimum margin

- Maximum discount exposure
- Maximum recommendation count
- Product exclusion
- Channel exclusion
- Customer exclusion
- Partner restrictions

Experience Guardrails

Prevent:

- Duplicate recommendations
- Recommending purchased products
- Irrelevant recommendations
- Excessive recommendations
- Incompatible products
- Unavailable products

Capacity Guardrails

Check:

- Ticket capacity
- Timeslot availability
- F&B availability
- Retail inventory
- Rental inventory
- Parking availability
- Experience capacity

Example

AI recommends:

VIP Experience Upgrade

Relevance: 94%

But remaining capacity:

0

Result:

SUPPRESSED — NO CAPACITY

The recommendation engine must never promote unavailable inventory simply because the AI score is high.

Screen 9 — Recommendation Policy, AI Control & Governance

Purpose

Define how much authority the TICVAI AI layer has over recommendation decisions.

AI Operating Modes

Mode 1 — Rules Only

No AI ranking.

Use manually configured relationships and rules.

Mode 2 — AI Recommendation

AI proposes recommendations.

Configured rules remain authoritative.

Mode 3 — AI Ranking

AI ranks approved eligible products.

Mode 4 — Governed AI Optimization

AI may automatically optimize ranking within approved boundaries.

AI Must Never Override

- Product eligibility
- Capacity
- Inventory

- Commercial restrictions
- Customer consent/preferences
- Contractual restrictions
- Promotion governance
- Venue rules

Confidence Threshold

Example:

AI confidence:

92%

Minimum required:

70%

Recommendation permitted.

Low Confidence

If confidence falls below threshold:

Use:

Rule-Based Fallback

Audit

Record:

- Recommendation generated
- Strategy used
- AI model/version
- Rule version
- Ranking factors
- Confidence
- Suppression reason
- Final recommendation

Screen 10 — Recommendation Strategy Simulator & AI Advisor

Purpose

Allow administrators to test recommendation strategies before activating them.

This should become the final validation screen for Board 1.

Simulation Inputs

Select:

- Customer profile
- Customer segment
- Basket
- Product
- Venue
- Channel
- Journey stage
- Date/time
- Membership
- Loyalty
- Inventory conditions

Example Scenario

Customer:

Family / Returning Guest

Current basket:

4 × Water Park Admission

Channel:

Mobile App

Journey:

Cart

Candidate Recommendations

1. Aquarium Admission
2. Family Meal
3. Photo Package
4. Parking
5. Annual Membership

Engine Result

#1 Aquarium

Relevance: 94%

Predicted acceptance: 31%

Expected incremental revenue: AED 280

#2 Family Meal

Relevance: 89%

Predicted acceptance: 44%

Expected incremental revenue: AED 160

#3 Photo Package

Relevance: 76%

Predicted acceptance: 22%

Expected incremental revenue: AED 70

Decision Trace

Show:

Candidate Products

↓

Eligibility

↓

Availability

↓

Guardrails

↓

Strategy Objective

↓

AI Score

↓

Ranking

↓

Final Recommendations

This makes the recommendation engine explainable.

AI Requirements for Board 1

AI is fundamental to this module, but it must operate within TICVAI's commercial rules.

AI shall support

- Product affinity analysis
- Recommendation scoring
- Propensity scoring
- Recommendation ranking
- Customer relevance
- Journey-context analysis
- Channel optimization
- Timing optimization

- Product relationship discovery
- Revenue prediction
- Margin prediction
- Acceptance prediction

AI shall identify

- New upsell relationships
 - New cross-sell relationships
 - High-performing touchpoints
 - Poor-performing recommendations
 - Recommendation fatigue
 - Missing product relationships
 - Capacity optimization opportunities
-

AI Recommendation Example

Opportunity Detected

Water Park → Aquarium

AI analysis:

Customers purchasing Water Park family tickets:

- 34% Aquarium affinity
- 27% historical cross-sell conversion
- Strongest response through Mobile App
- Best placement: Cart
- Best timing: Immediately after ticket selection

Recommendation

Enable Aquarium cross-sell for Family Water Park customers in Mobile App Cart.

Expected:

+AED 420K incremental annual revenue

Confidence:

88%

Authorized user can:

Simulate

↓

Create Strategy Draft

↓

Approve

↓

Activate

AI should not bypass the governance process.

Real-Time Recommendation Architecture

I recommend implement the recommendation flow as:

Customer / Basket / Context

↓

Recommendation API

↓

1. Candidate Product Generation

What could potentially be recommended?

↓

2. Eligibility

Can this customer buy it?

↓

3. Availability

Can TICVAI actually sell it now?

↓

4. Business Guardrails

Should TICVAI recommend it?

↓

5. AI / Rule Scoring

How relevant is it?

↓

6. Ranking

Which recommendation is strongest?

↓

7. Placement Rules

How many can this channel display?

↓

Final Recommendation Response

This service should be centralized rather than rebuilding recommendation logic separately in B2C, POS, Kiosk and Mobile.

Recommendation Response Example

The engine should return more than simply a Product ID.

Conceptually:

Recommendation

Aquarium Admission

Type

Cross-Sell

Rank

1

Reason

High product affinity

Confidence

92%

Placement

Cart

Price

AED 120

Promotion

FAMILYCOMBO eligible

Availability

Available

Tracking ID

REC-XXXXXX

The Tracking ID is important so TICVAI can later determine whether the recommendation generated a sale.

Relationship with Promotions & Bundles

The architecture should be:

Recommendation Engine

“Recommend Aquarium.”

↓

Promotion Engine

“This customer qualifies for 15% Aquarium discount.”

↓

Pricing Engine

“Final price = AED 102.”

↓

Recommendation UI

Add Aquarium for AED 102

This prevents the Recommendation Engine from becoming another pricing engine.

Relationship with Dynamic Pricing

Likewise:

Recommendation Engine:

Which product should we offer?

Dynamic Pricing:

What should that product cost?

Promotion Engine:

Which benefit applies?

These should remain independent services even though the customer experiences them as one offer.

Board 1 Roles

Recommended roles:

- Marketing Manager
 - Commercial Manager
 - Revenue Manager
 - CRM Manager
 - E-Commerce Manager
-

- Venue Manager
 - F&B Manager
 - Retail Manager
 - Membership Manager
 - B2B Manager
 - Data Analyst
 - System Administrator
-

Key Integrations

Ticketing Catalogue

Products and relationships.

Promotions & Bundles

Offers and bundles.

CRM

Customer profiles and segments.

Membership & Loyalty

Membership/tier behavior.

F&B

Menu/product recommendations.

Retail

Retail recommendations.

Inventory

Availability.

Resource / Capacity

Capacity availability.

Dynamic Pricing

Current price.

Payment

Checkout context.

B2C / Mobile

Digital recommendation placements.

POS / Kiosk

Frontline recommendations.

B2B / Partner

Commercial/channel recommendations.

Analytics / AI

Models, performance and optimization.

Scope Boundary with Future Boards

To avoid duplication across the six boards:

Board 1 — Command Center & Strategy

Controls the overall recommendation ecosystem.

Board 2 — Upsell & Upgrade Engine

Detailed upgrade/up-tier recommendation logic.

Board 3 — Cross-Sell & Product Affinity Engine

Detailed complementary-product intelligence.

Board 4 — Customer Journey & Contextual Recommendation Engine

Detailed journey, location and contextual orchestration.

Board 5 — AI Personalization & Next-Best-Offer Engine

Advanced customer-level AI decisioning.

Board 6 — Governance, Experimentation & Analytics

Testing, attribution, performance and optimization.

This separation will keep the 60 screens clean and avoid repeating the same recommendation functionality across boards.

Acceptance Criteria

Board 1 shall be complete when:

1. Administrators have a centralized Recommendation Command Center.
2. Recommendation strategies can be created and managed.
3. Strategies can have different commercial objectives.
4. Multi-objective optimization can be configured.
5. Recommendation relationships can be defined centrally.
6. Upsell, upgrade, cross-sell, add-on, bundle and membership relationships are supported.
7. Relationships can be manual, rule-generated or AI-discovered.
8. Recommendation placements can be configured.
9. Journey-stage-specific placements are supported.
10. Channel-specific behavior is configurable.
11. B2C, App, POS, Kiosk, B2B and API can consume the centralized recommendation engine.
12. Candidate recommendations can be ranked.
13. Ranking can use commercial and AI factors.
14. Maximum recommendations per placement can be configured.
15. Duplicate recommendations are prevented.
16. Already-purchased products are suppressed.
17. Unavailable products are suppressed.
18. Capacity constraints are respected.

19. Commercial guardrails are enforced.
20. AI operating mode can be configured.
21. AI confidence thresholds can be configured.
22. Rule-based fallback is supported.
23. AI cannot override authoritative eligibility/inventory/commercial controls.
24. Recommendation decisions are explainable.
25. Recommendation decisions can be audited.
26. Administrators can simulate strategies before activation.
27. Simulation shows candidate, rejected and final recommendations.
28. Recommendation tracking IDs support downstream attribution.
29. Recommendation APIs return the required product, ranking, reason and tracking context.
30. Pricing, promotion and recommendation responsibilities remain separated.
31. AI can discover potential product relationships.
32. AI can recommend strategy improvements.
33. AI recommendations can move into governed draft/approval workflows.
34. The engine can support future recommendation types without redesigning its core architecture.
35. All tenant data and configuration are isolated and governed through TICVAI RBAC/PBAC.

Board 1 is locked as: *Recommendation Command Center & Strategy* — 10 backend screens.

Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 2 — Upsell & Upgrade Recommendation Engine



Board Objective

Board 2 shall provide TICVAI with the specialized **Upsell & Upgrade Recommendation Engine** responsible for identifying when a customer can be moved from their current product, ticket, experience, membership, bundle, or service to a **higher-value or more valuable alternative**.

Board 1 established the overall recommendation strategy, placements, ranking, guardrails, and AI governance.

Board 2 answers:

“Given what the guest is currently considering or already owns, what better option should TICVAI recommend, why should it be recommended, and what incremental commercial value would the upgrade create?”

Examples include:

Standard Admission → Premium Admission

Single Attraction → Multi-Attraction Pass

General Admission → Fast Track

Day Ticket → Annual Membership

Standard Bundle → Premium Bundle

Basic Experience → VIP Experience

Existing Membership → Higher Membership Tier

The matrix specifically requires dynamic upselling during the booking journey, membership upgrades, higher-value products and experiences, AI-driven recommendations, post-purchase opportunities, and synchronized recommendations across channels.

Board 2 shall contain 10 backend screens.

Screen 1 — Upsell & Upgrade Command Center

Purpose

Provide centralized operational and commercial visibility into all upsell and upgrade activities.

KPI Cards

- Active Upsell Rules
- Active Upgrade Paths
- Upsell Recommendations
- Upgrade Offers Presented
- Acceptance Rate
- Upgrade Conversion
- Upsell Revenue
- Incremental Revenue
- Average Upgrade Value
- AOV Uplift

- Margin Uplift
- AI-Generated Opportunities

Upsell Performance by Type

Break down:

- Ticket Upgrade
- Experience Upgrade
- Bundle Upgrade
- Membership Upgrade
- Fast Track
- Premium Access
- Seating Upgrade
- Add-On Upgrade
- Service Upgrade

Performance Trend

Show:

Recommendations → Accepted → Purchased → Incremental Revenue

Health Status

Each upgrade strategy:

- Healthy
- Monitor
- Underperforming
- Inventory Limited
- Capacity Limited
- Conflict
- Suspended

AI Insight

Example:

Guests purchasing Standard Water Park admission are 3.2x more likely to accept the Premium Admission upgrade when the price difference is below AED 60.

Screen 2 — Upgrade Path & Product Ladder Builder

Purpose

Define valid upgrade relationships between products.

TICVAI should not allow the AI to invent arbitrary upgrade paths without governed product relationships.

Product Ladder Example

Standard Admission

↓

Premium Admission

↓

VIP Admission

Another Example

Single Attraction

↓

2 Attraction Pass

↓

All Attraction Pass

↓

Annual Membership

Configuration

For each relationship:

- Source product

- Target product
- Upgrade type
- Upgrade sequence
- Priority
- Business entity
- Venue
- Channel
- Effective period
- Status

Relationship Types

- Direct Upgrade
- Tier Upgrade
- Experience Upgrade
- Bundle Upgrade
- Membership Upgrade
- Seat Upgrade
- Service Upgrade

Multi-Step Paths

Allow:

Basic → Standard → Premium → VIP

Administrators may specify whether TICVAI can:

- Recommend next level only
- Skip levels
- Recommend multiple levels

Screen 3 — Upsell Eligibility & Qualification Rules

Purpose

Determine whether an upsell opportunity is valid for the current customer and transaction.

This screen provides **upgrade-specific qualification**, while the reusable customer targeting capabilities remain centralized in the recommendation/CRM eligibility architecture.

Conditions

Support:

- Current product
- Product variant
- Customer type
- Guest type
- Membership
- Loyalty
- Channel
- Venue
- Visit date
- Timeslot
- Quantity
- Basket value
- Purchase status
- Previous purchase
- Upgrade history

Example

IF

Current Product = Standard Admission

AND

Guest Type = Adult

AND

Premium Capacity > 0

AND

Channel = B2C

THEN

Premium Admission Upgrade = Eligible.

Exclusions

Prevent upgrade if:

- Target already purchased
 - Target included in bundle
 - Customer not eligible
 - Product expired
 - Upgrade deadline passed
 - Capacity unavailable
 - Contract prohibits upgrade
 - Ticket status incompatible
-

Screen 4 — Upgrade Price Difference & Value Proposition Manager

Purpose

Configure how TICVAI calculates and communicates the commercial difference between the current product and the recommended upgrade.

The Recommendation Engine should not become the authoritative pricing engine.

It should consume the authoritative price difference from Pricing/Promotion services.

Display

Current Ticket

AED 180

Premium Ticket

AED 240

Upgrade Difference

+ AED 60

Customer Message

Upgrade to Premium for only AED 60 more.

Value Proposition

Configure benefits such as:

- Fast Track
- Additional attractions
- Premium seating
- Extra validity
- Exclusive access
- Included meal
- Free parking
- Additional visits
- VIP experience

Value Templates

Example:

For AED 60 more, receive Fast Track + Meal + Premium Access.

Value Metrics

TICVAI may calculate:

Additional benefit value

versus

Upgrade price

to help rank upgrades.

Screen 5 — Ticket, Experience & Bundle Upgrade Manager

Purpose

Configure detailed upsell relationships for ticketing and experience products.

Ticket Upgrade Examples

- Standard → Premium
- General Admission → Fast Track
- Single Day → Multi-Day
- Single Attraction → Multi-Attraction
- Restricted → Flexible
- Off-Peak → Anytime

Experience Upgrade

- Standard Tour → Private Tour
- General Experience → VIP Experience
- Basic Activity → Premium Activity

Bundle Upgrade

Standard Family Bundle

↓

Premium Family Bundle

Comparison View

Benefit	Standard	Premium
Admission	✓	✓

Benefit	Standard	Premium
---------	----------	---------

n

Fast Track	—	✓
------------	---	---

Meal	—	✓
------	---	---

Parking	—	✓
---------	---	---

Photo	—	✓
-------	---	---

Upgrade Impact

Show:

Original Basket: **AED 720**

Premium Basket: **AED 940**

Incremental Revenue:

AED 220

Screen 6 — Membership & Pass Upgrade Engine

Purpose

Handle one of the most commercially valuable upgrade scenarios:

Converting visitors into members/pass holders.

Upgrade Scenarios

- Day Ticket → Annual Pass
- Single Attraction Pass → Multi-Attraction Pass
- Basic Membership → Silver
- Silver → Gold
- Gold → Platinum
- Individual → Family Membership

Break-Even Intelligence

Example:

Customer currently spends:

AED 420

Annual Membership:

AED 650

Difference:

AED 230

AI/customer message:

Upgrade for AED 230 more and receive unlimited admission for 12 months.

Visit-Based Recommendation

Example:

Customer has visited:

3 times in 90 days

TICVAI detects that annual membership may offer stronger customer value.

Membership Upgrade Inputs

- Historical visits
- Historical spend
- Current basket
- Membership tier
- Expiry
- Renewal status
- Household/family context where permitted
- Loyalty tier

Important Integration

Actual membership entitlement creation remains in the **Membership Module**.

Board 2 only recommends and initiates the upgrade journey.

Screen 7 — Pre-Purchase, Cart & Checkout Upsell Manager

Purpose

Control upselling during the active purchase journey.

Journey Stages

Product Page

Recommend higher product.

Availability Selection

Recommend premium timeslot/experience where applicable.

Ticket Selection

Recommend better ticket.

Cart

Recommend upgrade.

Checkout

Last permitted upgrade opportunity.

Example

Customer selects:

Standard Admission — AED 180

TICVAI displays:

Go Premium for AED 45 more

Benefits:

- ✓ Fast Track
- ✓ Meal Voucher
- ✓ Premium Zone

Placement Rules

Configure:

- Maximum upgrade cards
- Upgrade order
- Display style
- Price-difference threshold
- Minimum relevance
- Suppression behavior

Cart-Level Upgrade

Allow TICVAI to identify:

Four individual tickets could be replaced by a Family Premium Bundle at a better commercial/customer-value outcome.

But any price/discount decision remains governed by the relevant Pricing and Promotion engines.

Screen 8 — Post-Purchase & In-Journey Upgrade Manager

Purpose

Support upgrades after the initial transaction where the product and operational rules permit them.

This is explicitly important because upselling should not stop at checkout.

Post-Purchase Touchpoints

- Confirmation page
- Confirmation email
- Mobile App
- Digital ticket
- Pre-visit communication
- Customer account
- Call Center
- POS

- Venue service desk

Example

Customer purchased:

Standard Water Park Ticket

Three days before visit:

Upgrade to Premium and skip the queue for AED 50.

Upgrade Windows

Configure:

- Immediately after purchase
- Up to X hours before visit
- Until first redemption
- After first redemption where allowed
- Until product expiry

Operational Validation

Before presenting:

Check:

- Ticket status
- Visit date
- Redemption status
- Capacity
- Product validity
- Upgrade policy

Important

If ticketing policy says:

No upgrades after first admission

the Recommendation Engine must respect that policy.

Screen 9 — Upsell Ranking, Propensity & AI Opportunity Engine

Purpose

Use AI and commercial logic to identify the strongest upgrade opportunity for each customer/context.

Candidate Example

Customer currently holds:

Standard Water Park

Potential upgrades:

1. Premium Water Park
2. Two-Attraction Pass
3. Annual Membership
4. VIP Experience

Scoring Dimensions

TICVAI may score:

- Customer relevance
- Upgrade propensity
- Historical acceptance
- Price sensitivity
- Incremental revenue
- Margin
- Product affinity
- Membership propensity
- Capacity
- Commercial priority

Example

Upgrade	Propensity	Incremental Rev.	Score
Premium	42%	AED 60	91
2 Attraction	34%	AED 120	88
Membership	18%	AED 470	76
VIP	8%	AED 350	51

Final Recommendation

Premium Admission

Reason:

Highest combination of acceptance probability, customer relevance and commercial value.

AI Discovery

AI can identify new upgrade opportunities such as:

Customers buying Product A frequently purchase Product B within seven days. Consider creating a governed upgrade path.

But AI-discovered relationships should enter a **Draft/Review** state before becoming authoritative.

Screen 10 — Upgrade Simulator, Comparison & AI Advisor

Purpose

Allow business users to simulate an upsell strategy before activation.

Simulation Inputs

- Customer segment/profile
- Current product
- Quantity
- Basket

- Channel
- Venue
- Visit date
- Membership
- Loyalty
- Journey stage

Scenario

Customer:

Family / Returning

Current:

4 × Standard Admission

Basket:

AED 720

Candidate 1

Premium Admission

Upgrade difference:

AED 240

Predicted acceptance:

38%

Expected incremental value:

AED 91.20 per eligible basket

Candidate 2

Family Premium Bundle

Upgrade difference:

AED 180

Predicted acceptance:

51%

Expected incremental value:

AED 91.80

Candidate 3

Annual Family Membership

Upgrade difference:

AED 780

Predicted acceptance:

14%

Expected incremental value:

AED 109.20

AI Recommendation

Recommend Family Premium Bundle

Reason:

- Highest acceptance-adjusted value
 - Strong customer relevance
 - Capacity available
 - Margin above target
 - Better family benefit proposition
-

Upgrade Decision Trace

Screen 10 should visually explain:

Current Product

↓

Candidate Upgrade Paths

↓

Eligibility Check

↓

Product / Capacity Availability

↓

Pricing Difference

↓

Commercial Guardrails

↓

AI Propensity

↓

Ranking

↓

Final Upgrade Recommendation

This gives TICVAI full explainability.

AI Requirements for Board 2

AI shall support:

- Upgrade propensity scoring
- Membership conversion propensity
- Price sensitivity estimation
- Product upgrade affinity
- Upgrade timing optimization
- Journey-stage optimization
- Channel optimization
- Expected incremental revenue
- Expected margin impact

- Upgrade relationship discovery

AI should answer:

Which upgrade?

For which customer?

At what point?

Through which channel?

At what price difference?

With what predicted acceptance?

AI Price Sensitivity

This is particularly useful.

Example historical behavior:

+AED 30

Acceptance: 47%

+AED 50

Acceptance: 38%

+AED 75

Acceptance: 19%

AI can tell Commercial:

Upgrade acceptance drops significantly when the incremental price exceeds approximately AED 55 for this segment.

However, AI does **not independently change the product price**.

It provides insight to Pricing/Dynamic Pricing or recommends a governed commercial action.

Upsell vs Cross-Sell — Important Boundary

We should keep Board 2 very clear.

Upsell / Upgrade

Customer replaces or enhances the current product with a higher-value version.

Standard → Premium

Cross-Sell

Customer keeps Product A and adds complementary Product B.

Water Park + Family Meal

That detailed functionality belongs in:

Board 3 — Cross-Sell & Product Affinity Engine

This prevents Board 2 and Board 3 from becoming duplicates.

Relationship with Ticket Service Policies

The upgrade engine must consume TICVAI's existing ticket-service rules.

For example:

Can Upgrade?

Yes.

Upgrade Deadline

2 hours before visit.

Price Difference Treatment

Pay difference.

Existing Promotion

Recalculate.

Ticket Reissue

Required.

The Recommendation Engine discovers the opportunity.

The Ticketing/Service Policy layer determines whether the upgrade transaction is operationally permitted.

Relationship with Promotion Engine

Example:

Original ticket:

AED 180

Target upgrade:

AED 240

Base difference:

AED 60

Promotion engine determines:

Customer qualifies for:

Premium Upgrade Campaign — AED 15 off

Final upgrade difference:

AED 45

The recommendation UI may therefore show:

Upgrade to Premium for AED 45 more.

The Recommendation Engine should not calculate the promotional discount itself.

Relationship with Dynamic Pricing

If Premium Admission currently costs:

AED 260

instead of its normal AED 240 because of dynamic pricing, Board 2 must consume:

Current authoritative price = AED 260

and calculate/display the upgrade opportunity based on the price returned by the pricing architecture.

Again:

Recommendation decides WHAT.

Pricing decides PRICE.

Promotion decides BENEFIT.

Omnichannel Architecture

Upgrade opportunities should be exposed centrally to:

- B2C
- Mobile App
- POS
- Mobile POS
- Kiosk
- Call Center
- B2B where permitted
- Partner/API where permitted

Example:

A customer buys Standard online.

Later opens the Mobile App.

The engine may still identify:

Premium Upgrade Available

because the recommendation decision is centralized rather than isolated inside the original B2C transaction.

Key Integrations

Recommendation Board 1

Strategy, ranking and placement.

Ticketing Catalogue

Product/variant relationships.

Ticket Service Policies

Upgrade eligibility.

Membership

Membership conversion and tier upgrades.

Promotions

Upgrade campaigns.

Pricing / Dynamic Pricing

Authoritative price.

Capacity

Target-product availability.

CRM

Customer behavior.

Loyalty

Tier and behavior.

B2C / Mobile

Digital upsell.

POS / Kiosk

Frontline upsell.

Payment

Collect additional payment.

Finance

Revenue and margin.

Analytics

Performance measurement.

Roles & Permissions

Recommended roles:

- Commercial Manager
- Revenue Manager
- Product Manager
- Marketing Manager
- Membership Manager
- E-Commerce Manager
- Venue Manager
- B2B Manager
- Data Analyst
- System Administrator

Sensitive Actions

Permissions should separately govern:

- Create upgrade path
 - Activate upgrade path
 - Change priority
 - Change commercial threshold
 - Enable AI discovery
 - Approve AI-generated relationship
 - Enable post-purchase upgrade
 - Override product restriction
-

Board 2 Acceptance Criteria

Board 2 shall be considered complete when:

1. Administrators have an Upsell & Upgrade Command Center.
2. Product upgrade paths can be configured.
3. Multi-level product ladders are supported.
4. Next-level-only and skip-level upgrades can be configured.
5. Ticket upgrades are supported.
6. Experience upgrades are supported.
7. Bundle upgrades are supported.
8. Membership/pass upgrades are supported.
9. Seat/service upgrades can be supported where applicable.
10. Upgrade-specific eligibility rules can be configured.
11. Product and capacity availability are checked before recommendation.
12. Existing purchases and included benefits prevent duplicate recommendations.
13. Authoritative upgrade price differences are consumed from pricing services.
14. Promotion-adjusted upgrade prices can be displayed.
15. Customer-facing upgrade value propositions can be configured.
16. Product benefit comparisons are supported.
17. Pre-purchase upselling is supported.
18. Cart upselling is supported.
19. Checkout upselling is supported.
20. Post-purchase upgrades are supported where policy allows.
21. Upgrade windows can be configured.
22. Redemption status is respected.
23. Ticket-service upgrade policies are respected.
24. AI can score upgrade propensity.

- 25. AI can score membership conversion propensity.
- 26. AI can identify price-sensitivity patterns.
- 27. AI can rank multiple valid upgrade opportunities.
- 28. AI can discover potential new upgrade relationships.
- 29. AI-discovered relationships require governed review/activation.
- 30. Administrators can simulate upgrade scenarios before activation.
- 31. Simulation shows acceptance probability and expected incremental value.
- 32. Recommendation decisions provide explainable traces.
- 33. All supported TICVAI channels can consume centralized upgrade recommendations.
- 34. Upgrade decisions are tracked for downstream attribution.
- 35. Pricing, Promotions, Ticket Policies and Recommendation responsibilities remain architecturally separated.

Board 2 is locked as: *Upsell & Upgrade Recommendation Engine* — 10 backend screens.

Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 3 — Cross-Sell & Product Affinity Engine



Board Objective

Board 3 shall provide TICVAI with the specialized **Cross-Sell & Product Affinity Engine** responsible for identifying complementary products, services, experiences, and entitlements that should be recommended **in addition to** what the guest is already buying or owns.

The distinction from Board 2 must remain clear:

Board 2 — Upsell: Replace or enhance the current product with a higher-value alternative.

Board 3 — Cross-Sell: Keep the current product and recommend additional complementary products.

Examples:

Water Park Ticket → Family Meal

Aquarium Ticket → Photo Package

Attraction Ticket → Parking

Event Ticket → Merchandise

Family Admission → Locker Rental

Museum Ticket → Guided Tour

Ticket → Another Attraction

Experience → F&B + Retail

The matrix requires cross-selling across tickets, attractions, F&B, Retail, external products, multiple destinations and other services, supported by AI product affinity and historical purchasing patterns.

Board 3 shall contain 10 backend screens.

Screen 1 — Cross-Sell Command Center

Purpose

Provide centralized visibility into cross-sell relationships, recommendation performance, attach rates and commercial opportunities.

KPI Cards

- Active Cross-Sell Relationships
- AI-Discovered Relationships
- Cross-Sell Recommendations
- Recommendations Accepted
- Cross-Sell Conversion
- Cross-Sell Revenue
- Estimated Incremental Revenue
- Attach Rate
- Items per Transaction

- AOV Uplift
- Cross-Category Revenue
- Suppressed Recommendations

Cross-Sell Mix

Break down by:

- Ticket → Ticket
- Ticket → F&B
- Ticket → Retail
- Ticket → Experience
- Ticket → Parking
- Ticket → Locker
- Ticket → Photo
- Ticket → Rental
- Ticket → Membership
- F&B → Retail
- Retail → F&B
- Internal → External Product

Performance Trend

Show:

Recommendations → Acceptances → Purchases → Revenue

Health Status

Relationships classified:

- High Performing
- Healthy
- Monitor
- Underperforming

- Low Affinity
- Inventory Limited
- Capacity Limited
- Suspended

AI Insight

Example:

Families purchasing Water Park admission have a 46% affinity with Family Meal packages and a 31% affinity with Photo Packages.

Screen 2 — Cross-Sell Relationship Builder

Purpose

Allow administrators to manually define governed complementary-product relationships.

Relationship Structure

Source Product

Water Park Admission

↓

Relationship

Cross-Sell

↓

Target Product

Family Meal

Configuration

For every relationship:

- Source product
- Target product
- Relationship type

- Relationship strength
- Priority
- Business entity
- Venue
- Channel
- Effective dates
- Customer segment
- Status

Relationship Types

Support:

- Complementary
- Frequently Bought Together
- Experience Extension
- Convenience
- Ancillary
- Destination Extension
- Food Pairing
- Retail Pairing
- Service Add-On
- Partner Product

Relationship Direction

Relationships may be:

One-Way

Water Park → Sunscreen

or:

Two-Way

Aquarium ↔ Water Park

Relationship Source

Identify:

- Manual
 - Historical Data
 - AI Discovered
 - Campaign
 - Partner
 - Imported
-

Screen 3 — Product Affinity Matrix & Relationship Map

Purpose

Provide a visual representation of how strongly products and categories relate to one another.

Affinity Matrix

Example:

Source	F&B	Retail	Photography	Parking	Aquarium
Water Park	82	64	71	56	76
Aquarium	68	72	79	34	—
Museum	48	76	52	31	44

Affinity Levels

- Very High
- High
- Medium

- Low
- No Relationship

Relationship Graph

Visualize:

Water Park

↙ ↓ ↘

Family Meal — Photo — Aquarium

Filters

- Venue
- Product
- Category
- Segment
- Channel
- Season
- Date range

Drill Down

Selecting a relationship shows:

- Transactions analyzed
 - Co-purchase rate
 - Attach rate
 - Revenue
 - Confidence
 - Historical trend
 - AI explanation
-

Screen 4 — Frequently Bought Together & Basket Pattern Engine

Purpose

Analyze transaction baskets to discover products commonly purchased together.

Basket Pattern Example

Customers buying:

Water Park Admission

also purchased:

Product	Co-Purchase	Attach Rate
Family Meal	46%	41%
Photo Package	34%	29%
Locker	31%	27%
Aquarium	28%	24%
Parking	22%	19%

Pattern Types

Analyze:

- Product pairs
- Product triplets
- Category relationships
- Sequential purchases
- Same-visit purchases
- Pre-visit purchases
- Post-purchase additions

Minimum Data Threshold

Administrators configure:

- Minimum transactions
- Minimum affinity
- Minimum confidence
- Minimum attach rate

This prevents AI from creating relationships from insignificant sample sizes.

AI Discovery

Example:

38% of customers purchasing Premium Aquarium + Family Meal subsequently purchase a Photo Package.

Suggested new relationship:

Aquarium → Photo Package.

Screen 5 — Cross-Category Recommendation Manager

Purpose

Manage recommendations across TICVAI's different commercial modules.

This is especially important because TICVAI is not only a ticketing platform.

Supported Categories

Ticketing

- Admissions
- Events
- Experiences
- Attractions
- Passes

F&B

- Meals
- Drinks

- Meal packages
- Restaurants

Retail

- Merchandise
- Souvenirs
- Apparel
- Accessories

Ancillary

- Parking
- Locker
- Photo
- Rental
- Transport

Membership

- Membership
- Loyalty-related products

Example Relationships

Water Park Ticket

→ Family Meal

→ Locker

→ Photo

→ Parking

→ Retail Package

Category Controls

Administrators may define:

Ticket products can recommend F&B, Retail, Parking and Photo.

but:

Parking products cannot generate recommendations.

Screen 6 — Multi-Attraction, Destination & Partner Cross-Sell

Purpose

Support cross-selling beyond the original venue or attraction.

This addresses the matrix requirements around multi-destination and external product combinations.

Internal Cross-Destination Example

Customer buys:

Water Park

Recommend:

Aquarium

or:

Museum

External Products

Where integrations exist:

- Hotel
- Transport
- Tour
- Attraction
- Event
- Destination experience
- Third-party service

Partner Relationship Configuration

For each relationship:

- Partner
- External product
- Product identifier
- Availability source
- Pricing source
- Commission
- Revenue share
- Settlement model
- Channel permissions

Example

Museum Admission

↓

Recommended:

City Sightseeing Tour

Partner:

External Tour Operator

Availability Requirement

External products must only be recommended when TICVAI can establish their current sellability through the configured integration.

Screen 7 — Basket-Aware Cross-Sell & Duplicate Prevention

Purpose

Make cross-selling aware of the customer's complete current basket.

This is critical because basic recommendation engines often recommend something the customer already owns.

Basket Analysis

Before recommending, check:

- Products already selected
- Bundle components
- Existing entitlements
- Included benefits
- Quantity
- Guest count
- Membership benefits
- Current promotions

Example

Basket:

4 × Water Park

1 × Family Meal

1 × Parking

Potential recommendations:

Family Meal  Already Purchased

Parking  Already Purchased

Photo Package ✓

Locker ✓

Aquarium ✓

Bundle Awareness

If the guest purchased:

Premium Family Bundle

and the bundle already includes:

Parking

the system must not recommend Parking again.

Quantity Intelligence

If basket has:

4 guests

and:

1 locker supports 4 guests

TICVAI should not blindly recommend four lockers unless the product rules justify it.

Screen 8 — Availability, Inventory & Capacity-Aware Cross-Sell

Purpose

Ensure recommendations reflect actual operational availability.

Validation Sources

Check:

- Ticket inventory
- Attraction capacity
- Timeslot capacity
- F&B availability
- Retail inventory
- Parking availability
- Locker availability
- Rental inventory
- Experience capacity
- Partner availability

Example

AI score:

Family Meal — 94

But restaurant availability:

Unavailable

Result:

SUPPRESSED

Next recommendation:

Photo Package — Score 89

moves upward.

Low Inventory Behavior

Configure:

- Continue recommending
- Reduce ranking
- Suppress
- Recommend substitute

Capacity Optimization

The engine may also use recommendations to redirect demand.

Example:

Aquarium 14:00:

95% utilized

Aquarium 16:00:

48% utilized

Recommendation:

Add Aquarium at 16:00.

provided the customer's itinerary and product rules allow it.

Screen 9 — AI Cross-Sell Discovery, Scoring & Ranking Engine

Purpose

Use AI to discover and rank the strongest complementary products for the current customer/context.

Candidate Generation

AI may consider:

- Current product
- Basket
- Customer segment
- Purchase history
- Visit history
- Product affinity
- Channel
- Journey stage
- Time
- Venue
- Availability
- Margin
- Historical acceptance

Candidate Example

Current:

Water Park Family Admission

Candidates:

Product	Affinity	Acceptance	Incremental Rev.	Score
Family Meal	94	46%	AED 180	96
Aquarium	91	28%	AED 320	92
Photo	86	34%	AED 95	88

Product	Affinity	Acceptance	Incremental Rev.	Score
Locker	78	31%	AED 60	81
Retail	63	17%	AED 120	67

Ranking

AI combines:

- Affinity
- Propensity
- Revenue
- Margin
- Availability
- Commercial priority
- Journey relevance

Explanation

Example:

Family Meal ranked #1 because 46% of similar family customers purchase this product, current inventory is available, and the recommendation historically increases basket value by AED 180.

Screen 10 — Cross-Sell Simulator & AI Opportunity Advisor

Purpose

Allow administrators to test cross-sell strategies and AI recommendations before activation.

Simulation Inputs

Select:

- Customer profile

- Segment
- Current product
- Basket
- Venue
- Channel
- Journey stage
- Date/time
- Membership
- Loyalty
- Inventory scenario

Example Scenario

Customer:

Family / Returning Visitor

Basket:

4 × Water Park Admission

Channel:

Mobile App

Journey:

Cart

Candidate Results

#1 Family Meal

Affinity: **94%**

Predicted Acceptance: **46%**

Price: AED 180

Expected Incremental Value:

AED 82.80

#2 Aquarium

Affinity: **91%**

Predicted Acceptance: **28%**

Price: AED 320

Expected Incremental Value:

AED 89.60

#3 Photo Package

Affinity: **86%**

Predicted Acceptance: **34%**

Price: AED 95

Expected Incremental Value:

AED 32.30

Strategy Decision

If objective:

Maximum Acceptance

→ Family Meal.

If objective:

Maximum Expected Incremental Revenue

→ Aquarium.

This illustrates why Board 1's strategy objectives are important.

Decision Trace

Screen 10 should show:

Current Basket

↓

Potential Cross-Sell Products

↓

Relationship / Affinity

↓

Customer Eligibility

↓

Duplicate / Bundle Check

↓

Inventory & Capacity

↓

Commercial Guardrails

↓

AI Scoring

↓

Strategy Objective

↓

Final Ranking

↓

Recommended Products

This gives users full explainability.

AI Requirements for Board 3

AI shall support:

- Product affinity discovery
- Basket association analysis

- Frequently-bought-together analysis
- Sequential purchase analysis
- Cross-category affinity
- Customer-level propensity
- Contextual relevance
- Expected incremental revenue
- Expected margin
- Cross-sell ranking
- New relationship discovery

AI should answer:

Customers buying Product A usually need what else?

Which complementary product is most relevant?

Which product creates the highest expected incremental value?

Which recommendation is likely to be accepted?

At which point in the journey should the relationship be used?

AI Relationship Discovery Governance

AI should not automatically create live commercial relationships from correlation alone.

Example:

AI discovers:

Water Park → Sunscreen

Co-purchase: 29%

Confidence: 91%

Estimated annual opportunity: AED 280K

Status:

AI Suggested

Authorized user actions:

Review

→ **Simulate**

→ **Approve Relationship**

→ **Activate**

This protects TICVAI from weak or inappropriate automated correlations.

Product Affinity vs Customer Personalization

Another important boundary:

Board 3

Focuses primarily on:

Product A is commercially/behaviorally related to Product B.

Board 5 — Personalization & Next-Best-Offer

Will focus on:

For this specific customer, Product B is more appropriate than Product C.

Board 3 therefore creates the **product intelligence foundation** that Board 5 can personalize.

Relationship with Promotions

Example:

Cross-Sell Engine:

Recommend Aquarium.

Promotion Engine:

Customer qualifies for 15% cross-attraction promotion.

Pricing:

Original price:

AED 120

Promotional price:

AED 102

Customer sees:

Add Aquarium for AED 102

Board 3 does not calculate the discount.

Relationship with Bundles

If Water Park + Aquarium already exists as a governed bundle:

The engine may recommend:

Upgrade your basket to the Explorer Bundle

depending on the configured strategy.

However:

- Board 3 identifies the affinity.
- Promotions/Bundle Engine owns the bundle.
- Board 2 may treat replacing the basket with a higher-value bundle as an upgrade.

This avoids creating a second bundle engine.

Relationship with F&B and Retail

TICVAI should not duplicate F&B or Retail product configuration here.

Board 3 consumes:

F&B Catalogue

and

Retail Catalogue

and builds recommendation relationships against those products.

Example:

Ticketing

Water Park Admission

↓

Recommendation Engine

Cross-sell relationship

↓

F&B

Family Meal

The F&B module remains the authoritative source for the meal itself.

Omnichannel Requirements

Cross-sell recommendations should be available to:

- B2C Website
- Mobile App
- POS
- Mobile POS
- Kiosk
- B2B where appropriate
- Call Center
- API
- Partner channels where permitted

The same recommendation can have different presentation behavior.

B2C

Complete your visit.

POS

Suggested Add-On: Family Meal.

Kiosk

Large visual add-on card.

API

Structured recommendation response.

Recommendation Tracking

Every recommendation must have a unique tracking identifier.

Example:

REC-CS-847291

Track:

- Recommendation generated
- Product
- Target product
- Customer/session where permitted
- Channel
- Placement
- Rank
- Impression
- Click/select
- Add to cart
- Purchase
- Redemption
- Revenue

This data becomes essential for **Board 6 — Recommendation Analytics & Experimentation**.

Key Integrations

Board 1 — Recommendation Strategy

Objectives, ranking and placement.

Board 2 — Upsell Engine

Upgrade relationships.

Ticketing Catalogue

Ticket products.

F&B

Food products.

Retail

Retail products.

Membership

Membership products.

Inventory

Stock.

Capacity

Availability.

Promotions & Bundles

Commercial offers.

Pricing

Authoritative price.

CRM

Customer behavior.

B2B / Partner

External relationships.

POS / Digital Channels

Recommendation delivery.

Analytics / AI

Affinity models.

Roles & Permissions

Recommended roles:

- Commercial Manager
- Product Manager
- Marketing Manager
- E-Commerce Manager
- F&B Manager
- Retail Manager
- Venue Manager
- Revenue Manager
- B2B Manager
- Data Analyst
- System Administrator

Controlled Actions

Permissions should govern:

- Create relationship
- Edit relationship
- Activate relationship
- Approve AI relationship
- Enable partner relationship
- Change affinity threshold

- Change category permissions
 - Override suppression
 - Import relationships
-

Board 3 Acceptance Criteria

Board 3 shall be considered complete when:

1. Administrators have a Cross-Sell Command Center.
2. Cross-sell relationships can be manually configured.
3. One-way and two-way relationships are supported.
4. Cross-category relationships are supported.
5. Ticket → Ticket cross-sell is supported.
6. Ticket → F&B is supported.
7. Ticket → Retail is supported.
8. Ticket → Experience is supported.
9. Ticket → Parking/Locker/Photo/Rental is supported.
10. Internal multi-attraction cross-selling is supported.
11. External/partner cross-selling is supported where integrated.
12. Product-affinity matrices can be viewed.
13. Product relationship maps can be viewed.
14. Basket-pattern analysis is supported.
15. Frequently-bought-together analysis is supported.
16. Minimum statistical thresholds can be configured.
17. Existing basket products are excluded from inappropriate recommendations.
18. Bundle components are recognized to prevent duplicate recommendations.
19. Existing entitlements are considered.

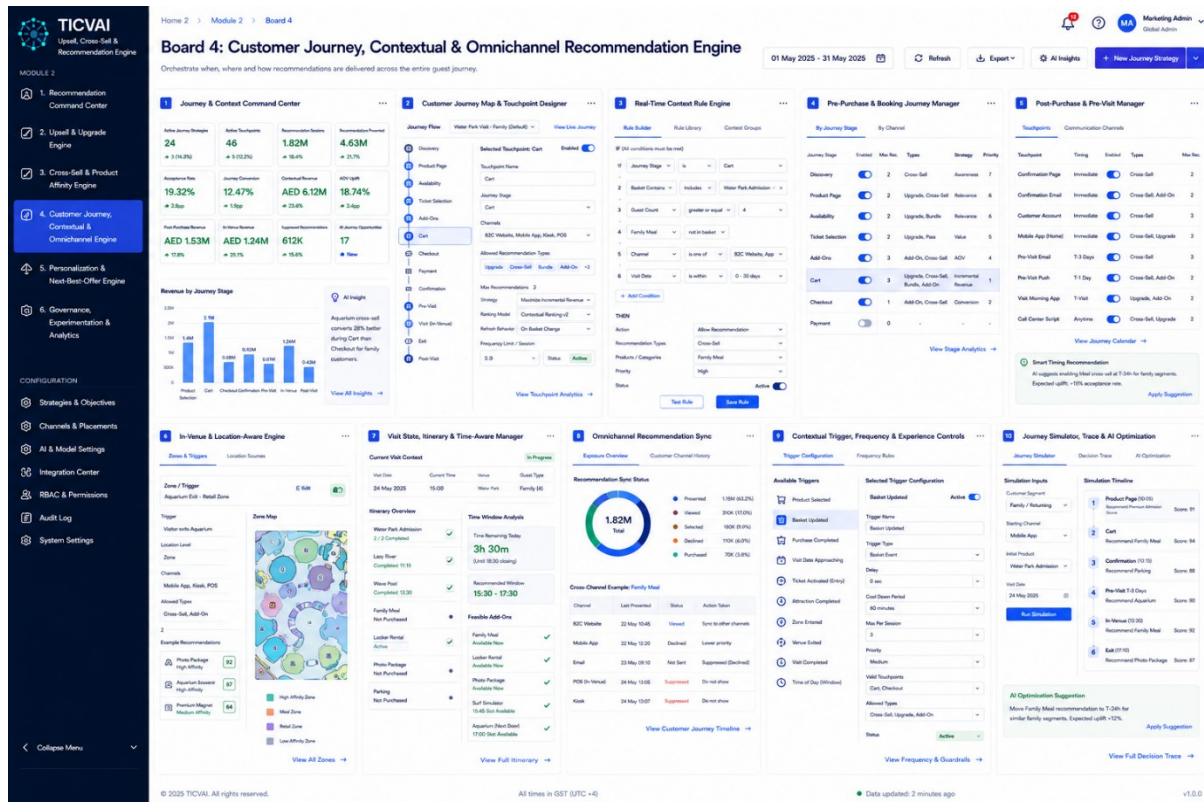
20. Quantity relationships can be respected.
21. Product inventory is checked before recommendation.
22. Attraction/timeslot capacity is checked.
23. F&B/Retail availability can affect recommendations.
24. Partner availability can affect recommendations.
25. Low availability can suppress or reduce recommendation rank.
26. AI can discover new product relationships.
27. AI can calculate affinity/propensity scores.
28. AI can rank complementary products.
29. AI recommendations provide explanations.
30. AI-discovered relationships require governed activation.
31. Administrators can simulate cross-sell scenarios.
32. Simulation can change according to commercial objective.
33. Recommendation tracking IDs are generated.
34. Recommendation interactions can be captured for attribution.
35. All supported channels consume the centralized Cross-Sell Engine rather than maintaining independent product-relationship logic.

Board 3 is locked as: *Cross-Sell & Product Affinity Engine — 10 backend screens.*

Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 4 — Customer Journey, Contextual & Omnichannel Recommendation Engine

Board Objective



Board 4 shall provide TICVAI with the centralized **Customer Journey & Contextual Recommendation Engine** responsible for determining **when, where, and in what context** an upsell, cross-sell, upgrade, bundle, membership, F&B, Retail, or ancillary recommendation should be presented.

Boards 2 and 3 answer:

WHAT should we recommend?

Board 4 answers:

WHEN and **WHERE** should we recommend it?

and:

What is happening in the guest's journey right now that should change the recommendation?

The same guest may need completely different recommendations depending on whether they are:

- Browsing an attraction
- Building a basket
- Checking out
- Waiting for their visit
- Entering the venue
- Inside the venue
- Leaving the venue
- Returning several weeks later

The matrix requires recommendations throughout the booking journey, post-purchase upsell/cross-sell, location-aware in-venue recommendations, multi-destination recommendations, CRM/context-aware offers, and synchronization across customer-facing channels.

Board 4 shall contain 10 backend screens.

Screen 1 — Journey & Context Command Center

Purpose

Provide centralized visibility into recommendation activity across the complete customer journey.

KPI Cards

- Active Journey Strategies
- Active Touchpoints
- Recommendation Sessions
- Recommendations Presented
- Acceptance Rate

- Journey Conversion
- Contextual Revenue
- Post-Purchase Revenue
- In-Venue Revenue
- AOV Uplift
- Suppressed Recommendations
- AI Journey Opportunities

Journey Funnel

Display:

Discovery

→ **Product**

→ **Selection**

→ **Cart**

→ **Checkout**

→ **Confirmation**

→ **Pre-Visit**

→ **In-Venue**

→ **Exit**

→ **Post-Visit**

Revenue by Journey Stage

Example:

Stage	Revenue
Product	AED
Selection	1.4M
Cart	AED
	2.1M

Stage	Revenue
-------	---------

Checkout	AED 680K
----------	-------------

Post-Purchase	AED 920K
---------------	-------------

Pre-Visit	AED 610K
-----------	-------------

In-Venue	AED 1.2M
----------	-------------

Post-Visit	AED 430K
------------	-------------

AI Insight

Aquarium cross-sell converts 28% better during Cart than Checkout for family customers.

Screen 2 — Customer Journey Map & Touchpoint Designer

Purpose

Allow TICVAI administrators to visually configure the customer journey and recommendation-enabled touchpoints.

Journey Designer

A visual flow:

Discovery

↓

Product Page

↓

Availability

↓

Ticket Selection

↓

Add-Ons

↓

Cart

↓

Checkout

↓

Payment

↓

Confirmation

↓

Pre-Visit

↓

Visit

↓

Post-Visit

Touchpoint Configuration

For each node configure:

- Recommendations enabled
- Recommendation types
- Maximum recommendations
- Strategy
- Placement
- Channel
- Ranking model
- Suppression policy

Example

Cart

Enabled: ✓

Maximum: 3

Allowed:

- Upgrade
- Cross-Sell
- Bundle

Strategy:

Maximize Incremental Revenue

Screen 3 — Real-Time Context Rule Engine

Purpose

Determine recommendation behavior using the customer's current transactional and operational context.

Context Inputs

Support:

- Current product
- Current basket
- Basket value
- Guest count
- Guest type
- Current venue
- Selected date
- Selected timeslot
- Current channel

- Journey stage
- Membership
- Loyalty
- Previous purchase
- Visit status
- Time until visit

Example

IF

Journey = Cart

AND

Basket contains Water Park

AND

Guest Count ≥ 4

AND

Family Meal not purchased

THEN

Allow Family Meal recommendation.

Context Groups

Support:

Transaction Context

Journey Context

Customer Context

Venue Context

Time Context

Operational Context

Nested Logic

Support:

- AND
 - OR
 - NOT
 - Include
 - Exclude
-

Screen 4 — Pre-Purchase & Booking Journey Recommendation Manager

Purpose

Manage recommendation behavior before transaction completion.

Supported Stages

Discovery

Recommend:

- Attractions
- Events
- Experiences

Product Page

Recommend:

- Premium alternatives
- Related attractions

Availability

Recommend:

- Alternative dates
- Alternative times
- Premium timeslots

Ticket Selection

Recommend:

- Upgrade
- Pass
- Membership

Cart

Recommend:

- F&B
- Retail
- Parking
- Photo
- Locker
- Other attractions

Checkout

Recommend only high-value, low-friction additions.

Stage-Specific Controls

Configure:

- Recommendation count
- Recommendation category
- Ranking
- Display priority
- Frequency
- Suppression

Customer Experience Rule

Avoid excessive selling.

Example:

Product Page: maximum 2

Cart: maximum 3

Checkout: maximum 1

Screen 5 — Post-Purchase & Pre-Visit Recommendation Manager

Purpose

Continue the commercial journey after payment while respecting the guest's existing purchase and visit plan.

Touchpoints

- Confirmation Page
- Confirmation Email
- Customer Account
- Mobile App
- Digital Ticket
- Wallet
- Pre-Visit Email
- Pre-Visit Push Notification
- Call Center

Example

Customer purchased:

Water Park — Saturday

Immediately after purchase:

Add Parking.

Three days before visit:

Add Family Meal.

Visit morning:

Add Fast Track if capacity is available.

Timing Rules

Configure:

- Immediately after purchase
- X days before visit
- X hours before visit
- Visit morning
- Before first redemption

Recommendation Sequence

Example:

T-7 Days

Aquarium.

T-3 Days

Meal.

T-1 Day

Parking.

T-2 Hours

Fast Track.

Suppression

Do not recommend if:

- Already purchased
 - Already included
 - Customer declined
 - Capacity unavailable
 - Upgrade window closed
-

Screen 6 — In-Venue & Location-Aware Recommendation Engine

Purpose

Support context-aware recommendations while the guest is physically visiting the venue.

This directly addresses the matrix's requirement for **location-aware in-venue upsell/cross-sell**.

Context Sources

Where technically available and permitted:

- Access-control scans
- Mobile App context
- Venue zone
- Attraction entry
- Ticket redemption
- Time
- Remaining entitlements
- Visit itinerary

Example

Guest enters:

Water Park

At 12:30:

Current context:

Family group.

Meal not purchased.

Nearby F&B location available.

Recommendation:

Family Lunch Package available nearby.

Zone Configuration

Example:

Aquarium Exit

Recommend:

- Photo
- Retail
- Next attraction

Theme Park Zone A

Recommend:

- F&B
- Fast Pass
- Nearby experience

Location Precision

Support configurable context levels such as:

- Venue
- Attraction
- Zone
- Access point

The recommendation engine should consume approved location/context signals rather than create a separate tracking system.

Screen 7 — Visit State, Itinerary & Time-Aware Recommendation Manager

Purpose

Make recommendations aware of **what the customer has already done and what remains possible during the visit.**

Visit States

- Not Arrived

- Arrived
- Entered
- In Venue
- Attraction Completed
- Partially Redeemed
- Exited
- Visit Completed

Itinerary Context

Show:

- Purchased products
- Reserved timeslots
- Completed activities
- Remaining activities
- Available time
- Venue closing time

Example

Current time:

15:00

Guest has:

Aquarium booking at **17:00**

Available window:

2 hours

Recommendation:

Add Museum Experience at 15:30.

Feasibility Check

Before recommending, verify:

- Travel time
- Duration
- Timeslot
- Capacity
- Venue closing
- Existing reservation

Important

AI should not recommend an experience that the guest cannot realistically complete.

Screen 8 — Omnichannel Recommendation Synchronization

Purpose

Coordinate recommendations across channels so customers receive a consistent experience.

Supported Channels

- B2C Website
- Mobile App
- POS
- Mobile POS
- Kiosk
- Call Center
- B2B
- API
- Partner channels where allowed

Customer Journey Example

Customer sees:

Aquarium recommendation on B2C

but does not purchase.

Later opens:

Mobile App

The engine determines whether to:

- Show again
- Change recommendation
- Suppress
- Reduce priority

Cross-Channel State

Track:

- Presented
- Viewed
- Selected
- Declined
- Purchased
- Suppressed
- Expired

Example

Recommendation:

Family Meal

B2C:

Presented.

App:

Presented.

Customer:

Declined twice.

POS:

Suppress Family Meal.

This prevents recommendation fatigue.

Channel Handoff

Context should move with the customer where identity/session linkage permits.

Screen 9 — Contextual Trigger, Frequency & Experience Controls

Purpose

Control exactly when a recommendation is triggered and prevent excessive customer interaction.

Trigger Types

- Product selected
- Basket updated
- Purchase completed
- Visit approaching
- Ticket activated
- Venue entered
- Attraction completed
- Zone entered
- Product redeemed
- Visit completed

Frequency Controls

Configure:

- Maximum recommendations/session
- Maximum/day
- Maximum/visit

- Maximum/channel
- Maximum per product
- Cooling period
- Repeat threshold

Decline Controls

Example:

Customer declines Photo Package:

1st time

May recommend later.

2nd time

Reduce rank.

3rd time

Suppress for current visit.

Experience Guardrails

Avoid:

- Recommendation overload
- Repeating rejected offers
- Bad timing
- Inappropriate journey interruption
- Recommendations immediately after complaints/service issues where such context is available and governed

Trigger Priority

If several triggers occur simultaneously:

TICVAI determines which recommendation event has priority.

Screen 10 — Journey Simulator, Decision Trace & AI Optimization

Purpose

Allow administrators to simulate an entire customer journey and see how recommendations change over time.

Simulation Profile

Customer:

Family / Returning Visitor

Channel:

Mobile App

Initial product:

Water Park

Visit:

Saturday

Simulated Journey

Stage 1 — Product Page

Recommendation:

Premium Admission

Reason:

High upgrade propensity.

Stage 2 — Cart

Recommendation:

Family Meal

Reason:

46% product affinity.

Stage 3 — Confirmation

Recommendation:

Parking

Reason:

Not included in basket.

Stage 4 — T-3 Days

Recommendation:

Aquarium

Reason:

Cross-attraction affinity.

Stage 5 — Venue Entry

Recommendation:

None.

Reason:

Experience guardrail — avoid immediate commercial message after entry.

Stage 6 — 12:30 In Venue

Recommendation:

Family Meal

Reason:

Meal not purchased + appropriate visit time + availability.

Stage 7 — Water Park Exit

Recommendation:

Photo Package

Reason:

Visit completed + photo available.

Journey Decision Trace

Screen 10 should visually show:

Customer Context

↓

Journey Stage

↓

Trigger

↓

Candidate Recommendations

↓

Eligibility

↓

Basket / Ownership Check

↓

Availability

↓

Frequency & Experience Controls

↓

AI Ranking

↓

Final Recommendation

This allows administrators to understand why the engine behaved differently at each stage.

AI Journey Optimization

AI shall analyze historical recommendation performance to determine:

- Best journey stage
- Best timing
- Best channel
- Best frequency
- Best recommendation sequence
- Best trigger
- Best recommendation count

Example

AI discovers:

Parking recommendations perform best immediately after purchase.

while:

F&B recommendations perform best 24–48 hours before the visit.

and:

Photo packages perform best after attraction redemption.

The engine can recommend different placement strategies accordingly.

AI Sequence Optimization

AI should also analyze **recommendation sequence**, not only individual offers.

Example:

Sequence A

Aquarium → Meal → Photo

Conversion:

21%

Sequence B

Meal → Aquarium → Photo

Conversion:

27%

AI Recommendation:

Use Sequence B for Family segments.

This becomes particularly valuable in longer guest journeys.

Journey Personalization Boundary

Board 4 handles:

What is happening in the journey right now?

Board 5 will handle:

Who exactly is this customer, and what is individually most relevant to them?

Example:

Board 4 knows:

Customer is currently in the Cart.

Board 5 knows:

This particular customer strongly prefers premium experiences and rarely purchases F&B.

Together:

Board 4 determines the right **moment**.

Board 5 determines the strongest **personalized offer**.

Location Privacy & Governance

Because this board may consume location/context signals, TICVAI should use only approved signals according to applicable configuration and consent requirements.

Principles

- Do not expose unnecessary precise location.
- Use venue/zone context where sufficient.
- Respect customer communication preferences.
- Respect Mobile App permissions.
- Retain only required recommendation events.
- Audit context usage where required.

Board 4 should **consume location events** from the appropriate application/access/venue layer rather than becoming a separate customer-location tracking platform.

Relationship with Access Control

This is particularly useful for TICVAI.

Example:

Access Control Event

Customer successfully entered Aquarium.

↓

Journey Engine

Updates:

Aquarium = Completed.

↓

Recommendation Engine

Stops recommending Aquarium.

↓

Later:

Recommends:

Retail / Photo / Next Attraction

Access Control therefore provides operational context but remains authoritative for admissions.

Relationship with Wallet / Digital Ticket

The digital ticket or wallet can become a recommendation touchpoint.

Example:

Ticket screen:

Water Park — Active

Below:

Complete your visit: Add Family Meal.

However, the Digital Ticket module should only **render the recommendation**.

Board 4 determines whether and when it appears.

Relationship with Notifications

Board 4 decides:

Recommendation opportunity exists at T–24 hours.

The communications/notification layer determines:

How the approved message is delivered.

This prevents Recommendation Engine from becoming another marketing communications platform.

Real-Time Context Architecture

I recommend:

Customer Event

Product Selected / Cart Updated / Purchase / Entry / Redemption

↓

Journey Context Service

Determine current journey state.

↓

Recommendation Engine

Generate candidates.

↓

Eligibility

↓

Availability

↓

Experience / Frequency Guardrails

↓

Board 5 Personalization

↓

Ranking

↓

Recommendation Response

↓

B2C / App / POS / Kiosk / API

Event Requirements

TICVAI should define standardized recommendation events such as:

- PRODUCT_VIEWED
- PRODUCT_SELECTED

- CART_UPDATED
- CHECKOUT_STARTED
- PURCHASE_COMPLETED
- TICKET_ACTIVATED
- VENUE_ENTERED
- ATTRACTION_REDEEMED
- VENUE_EXITED
- VISIT_COMPLETED
- RECOMMENDATION_PRESENTED
- RECOMMENDATION_SELECTED
- RECOMMENDATION_DECLINED
- RECOMMENDATION_PURCHASED

This event architecture will be important for both real-time decisioning and analytics.

Key Integrations

Board 1

Recommendation strategy and placement.

Board 2

Upsell candidates.

Board 3

Cross-sell candidates.

Board 5

Personalization.

Ticketing

Products, bookings and visit dates.

Access Control

Entry/redemption events.

CRM

Customer context.

Membership & Loyalty

Customer state.

Promotions

Applicable commercial offers.

Capacity / Inventory

Real-time sellability.

F&B / Retail

Product availability.

Mobile App

Digital and location-aware experiences.

Wallet / Ticket Media

Post-purchase touchpoints.

Notifications

Approved outbound communication.

Analytics / AI

Journey optimization.

Roles & Permissions

Recommended roles:

- Marketing Manager
- E-Commerce Manager
- Commercial Manager

- CRM Manager
- Venue Manager
- Guest Experience Manager
- Revenue Manager
- Digital Product Manager
- Data Analyst
- System Administrator

Sensitive Permissions

Separately control:

- Enable in-venue triggers
 - Enable location-aware context
 - Configure customer frequency
 - Enable post-purchase recommendation
 - Change journey strategy
 - Enable cross-channel synchronization
 - Approve AI journey optimization
-

Board 4 Acceptance Criteria

Board 4 shall be considered complete when:

1. TICVAI has a centralized Journey & Context Command Center.
2. Administrators can visually configure recommendation touchpoints.
3. Pre-purchase stages are supported.
4. Product-selection recommendations are supported.
5. Cart recommendations are supported.
6. Checkout recommendations are supported.
7. Post-purchase recommendations are supported.

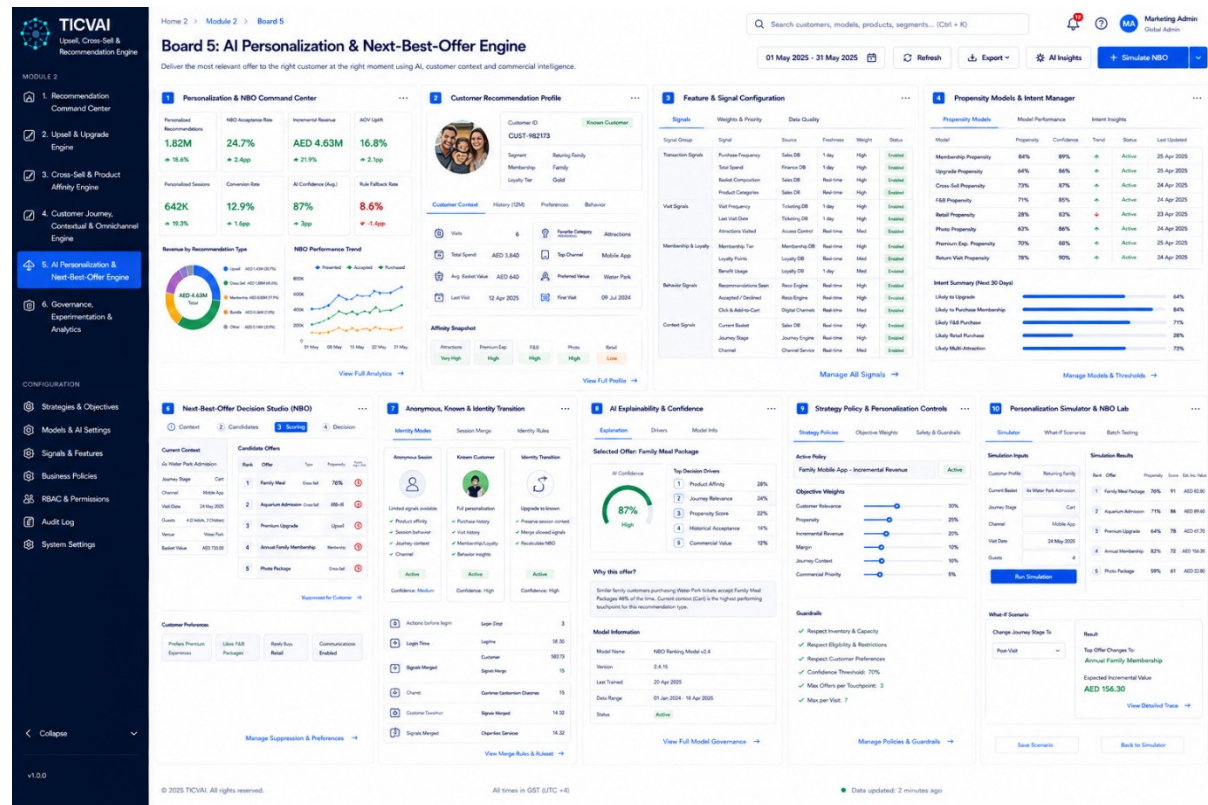
8. Pre-visit recommendations are supported.
9. In-venue recommendations are supported where context is available.
10. Post-visit recommendations are supported.
11. Context rules support current product and basket.
12. Context rules support channel.
13. Context rules support journey stage.
14. Context rules support visit date/time.
15. Visit state can affect recommendations.
16. Access-control/redemption events can update journey state.
17. Existing itinerary is considered.
18. Time feasibility can be checked before recommending an experience.
19. Availability and capacity remain authoritative.
20. Recommendations can synchronize across supported channels.
21. Previous recommendation exposure is considered.
22. Previous declines can reduce ranking or suppress recommendations.
23. Previous purchases suppress duplicate recommendations.
24. Recommendation frequency limits are configurable.
25. Cooling periods are configurable.
26. Trigger priority can be configured.
27. Customer-experience guardrails are supported.
28. Administrators can simulate a complete recommendation journey.
29. Simulation explains each recommendation decision.
30. AI can determine effective recommendation timing.
31. AI can determine effective journey stages.
32. AI can optimize recommendation sequences.
33. AI can identify recommendation fatigue.

- 34. Location-aware behavior respects configured privacy/permission requirements.
- 35. The Journey Engine consumes operational events from authoritative TICVAI modules rather than duplicating their functionality.
- 36. Standardized recommendation/customer-journey events are available for downstream analytics.
- 37. The architecture supports real-time recommendation decisioning.
- 38. B2C, App, POS, Kiosk and authorized APIs can consume the same centralized journey context.
- 39. Journey context and customer personalization remain separated but interoperable.
- 40. All configuration and decision activity respects TICVAI tenant isolation, RBAC/PBAC and audit requirements.

Board 4 is locked as: *Customer Journey, Contextual & Omnichannel Recommendation Engine — 10 backend screens.*

Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 5 — AI Personalization & Next-Best-Offer Engine



Board Objective

Board 5 shall provide TICVAI with the **customer-level AI personalization and Next-Best-Offer (NBO) decision engine**.

Boards 1–4 established:

- **Board 1:** overall recommendation strategy.
- **Board 2:** possible upsells and upgrades.
- **Board 3:** complementary products and product affinity.
- **Board 4:** when and where recommendations should appear.

Board 5 answers the most advanced question:

“For this specific guest, among all valid opportunities available right now, what is the most relevant and commercially appropriate offer TICVAI should present?”

The engine should combine customer information, transaction history, visit behavior, membership/loyalty information, current basket, real-time context, product affinity, historical response, commercial objectives, and operational availability.

It must support personalization without allowing AI to override authoritative pricing, eligibility, capacity, consent, or commercial governance.

Board 5 shall contain 10 backend screens.

Screen 1 — Personalization & NBO Command Center

Purpose

Provide centralized visibility into the operation and commercial impact of personalized recommendations.

KPI Cards

- Personalized Recommendations
- Personalized Customers/Sessions
- Next-Best-Offers Generated
- NBO Acceptance Rate
- Personalized Conversion
- Incremental Revenue
- Personalized AOV Uplift
- Recommendation Relevance
- AI Confidence
- Rule-Based Fallback Rate
- Suppressed Recommendations
- AI Opportunities

Decision Mix

Show recommendations generated by:

- Rules
- Product Affinity
- Segment Model
- Customer Propensity
- Next-Best-Offer AI
- Hybrid AI + Rules

Recommendation Categories

- Upsell
- Cross-Sell
- Membership
- Bundle
- F&B
- Retail
- Experience
- Ancillary
- Return Visit

AI Insight

Returning family visitors with two or more attraction purchases show a 3.1x higher propensity for multi-attraction membership than first-time family visitors.

Screen 2 — Customer Recommendation Profile

Purpose

Provide the recommendation engine's **commercially relevant customer context** used for personalization.

This should not become another CRM profile screen.

CRM remains the authoritative source of customer information.

Board 5 displays only the attributes relevant to recommendation decisioning.

Profile Sections

Customer Context

- Customer ID
- Known / Anonymous
- Customer segment
- Membership
- Loyalty tier
- Customer lifecycle stage
- Preferred channel where available

Commercial History

- Historical spend
- AOV
- Purchase frequency
- Product categories
- Visits
- Recent purchases
- Upgrade history
- Cross-sell acceptance

Recommendation Behavior

- Recommendations presented
- Accepted
- Declined
- Ignored

- Purchased
- Suppressed

Example

Customer Profile

Returning Family

Visits: 6

12-Month Spend: **AED 3,840**

Strong Affinity:

- Attractions
- Premium Experiences
- Photo

Low Affinity:

- Retail

Membership Propensity:

84%

Screen 3 — Customer Feature & Signal Configuration

Purpose

Control which approved data signals the AI may use when calculating personalized recommendations.

This is critical for governance and explainability.

Signal Groups

Transaction Signals

- Purchase frequency
- Basket value
- Products purchased

- Product categories
- Spend
- Purchase recency

Visit Signals

- Visit frequency
- Attractions visited
- Redemption history
- Visit duration where available
- Repeat visitation

Membership & Loyalty

- Membership tier
- Loyalty tier
- Points
- Benefits
- Membership expiry

Recommendation Signals

- Previous recommendation acceptance
- Declines
- Click/select behavior
- Recommendation frequency

Current Context

- Basket
- Channel
- Journey stage
- Venue
- Visit date

- Current entitlements

Signal Controls

For each signal:

- Enabled
 - Disabled
 - Source
 - Freshness
 - Weight
 - Availability
 - Data-quality status
-

Screen 4 — Propensity Model & Customer Intent Manager

Purpose

Calculate the likelihood that a guest will take specific commercial actions.

Propensity Models

Examples:

- Upgrade Propensity
- Membership Propensity
- Cross-Sell Propensity
- F&B Purchase Propensity
- Retail Purchase Propensity
- Bundle Propensity
- Premium Experience Propensity
- Return Visit Propensity

Example

Customer:

Returning Family

Propensity Score

Membershi
p 84%

Aquarium 77%

Family
Meal 71%

Photo 63%

Retail 28%

Propensity Bands

- Very High
- High
- Medium
- Low
- Very Low

Model Information

Show:

- Model name
- Model version
- Last updated
- Training period
- Confidence
- Population
- Data freshness

Important

Propensity is a **decision input**, not the final decision.

A 95% propensity does not mean the product should automatically be recommended if inventory, eligibility, journey, or commercial controls reject it.

Screen 5 — Next-Best-Offer Decision Studio

Purpose

This is the core screen of Board 5.

Combine all eligible candidates and determine the strongest personalized recommendation.

Candidate Sources

Candidates can come from:

Board 2

Upsells.

Board 3

Cross-sells.

Promotions

Eligible promotional offers.

Bundles

Available bundles.

Membership

Membership opportunities.

Journey Engine

Contextually appropriate opportunities.

Candidate Example

Customer currently buying:

4 × Water Park Tickets

Candidates:

Candidate	Type	Propensity	Commercial Value	Context
Family Meal	Cross-Sell	76%	Medium	94
Aquarium	Cross-Sell	71%	High	91
Premium Upgrade	Upsell	64%	Medium	88
Annual Membership	Membership	82%	Very High	72
Photo	Cross-Sell	59%	Low	84

NBO Result

Family Meal

Reason:

Highest combination of customer propensity, current journey relevance, product affinity and acceptance probability.

Next Alternative

Aquarium

The engine should be able to return a ranked list, not only one offer.

Screen 6 — Personalized Ranking & Decision Policy Builder

Purpose

Configure how TICVAI converts AI scores and commercial factors into the final recommendation ranking.

Ranking Dimensions

Support:

- Customer propensity
- Product affinity
- Journey relevance
- Historical acceptance

- Revenue
- Incremental revenue
- Margin
- Capacity
- Inventory
- Commercial priority
- Campaign priority
- Customer experience

Weighted Strategy Example

Customer Relevance — 30%

Propensity — 25%

Incremental Revenue — 20%

Margin — 10%

Journey Context — 10%

Commercial Priority — 5%

Different Policies

Policies may differ by:

- Tenant
- Venue
- Channel
- Customer segment
- Journey stage
- Recommendation category

Example

POS may optimize:

Speed + relevance.

Mobile App may optimize:

Personalization + incremental value.

Screen 7 — Customer Preference, Fatigue & Suppression Intelligence

Purpose

Prevent personalization from becoming repetitive or intrusive.

Board 4 provides general journey frequency controls.

Board 5 adds **customer-specific behavioral suppression**.

Customer Signals

Track:

- Repeated declines
- Repeated ignores
- Recent purchase
- Previous acceptance
- Category preference
- Recommendation fatigue
- Channel response

Example

Customer declined:

Photo Package

3 times.

System action:

Suppress Photo recommendation for 30 days.

Preference Learning

Example:

Customer consistently accepts:

- Premium experiences
- Fast Track
- F&B

and consistently ignores:

- Retail

The AI may reduce Retail ranking.

Controls

Configure:

- Decline threshold
- Ignore threshold
- Cooling period
- Category suppression
- Product suppression
- Channel suppression

Explicit Preferences

Where customers explicitly configure communication or recommendation preferences, those preferences must take precedence over inferred AI behavior.

Screen 8 — Anonymous, Known & Identity-Transition Personalization

Purpose

Allow useful recommendations even when TICVAI does not yet know the customer's identity.

Anonymous Session

Available signals may include:

- Current product
- Basket
- Channel
- Journey
- Session behavior
- Product affinity
- Venue
- Date/time

Recommendation:

Contextual / affinity-based

Known Customer

Additional signals:

- Purchase history
- Membership
- Loyalty
- Visits
- Previous recommendation behavior

Recommendation:

Personalized

Identity Transition

Example:

Guest browses anonymously.

↓

Adds Water Park.

↓

Logs in.

TICVAI should transition from:

Session-based recommendations

to:

Customer-aware recommendations

without unnecessarily repeating previously displayed offers.

Merge Rules

Configure how permitted session signals are associated with the known customer context.

Screen 9 — AI Explainability, Confidence & Model Governance

Purpose

Make personalized AI decisions understandable and governable.

Recommendation Explanation

Example:

Recommended

Annual Family Membership

Why?

- Customer visited 5 times in 8 months.
- Historical spend is AED 1,840.
- Current family basket is AED 720.
- Similar customers show strong membership conversion.
- Membership is currently available.
- Recommendation is permitted at this journey stage.

Confidence

87% — High

Decision Drivers

Show ranked contributors:

1. Visit Frequency
2. Family Segment
3. Historical Spend
4. Product Affinity
5. Current Basket

Model Governance

Display:

- Model
- Version
- Deployment date
- Owner
- Status
- Confidence threshold
- Fallback policy

Model States

- Draft
- Testing
- Approved
- Active
- Monitor
- Suspended
- Retired

AI Override

Authorized users may:

- Disable model

- Change approved threshold
- Switch to rules
- Suspend recommendation category

All actions must be audited.

Screen 10 — Personalization Simulator & Next-Best-Offer Lab

Purpose

Allow administrators to simulate how TICVAI would personalize recommendations for different customers before deploying changes.

Simulation Inputs

Select:

- Customer or synthetic profile
- Segment
- Membership
- Loyalty
- Historical spend
- Visit frequency
- Purchase history
- Current basket
- Venue
- Channel
- Journey stage
- Date/time

Example Customer

Returning Family

Visits:

5

12-month spend:

AED 2,900

Current basket:

4 × Water Park

Channel:

Mobile App

Stage:

Cart

Candidate Results

Family Meal

Propensity: 81%

Affinity: 92%

Context: 96%

NBO Score:

91

Aquarium

Propensity: 74%

Affinity: 89%

Context: 91%

NBO Score:

86

Annual Membership

Propensity: 88%

Commercial Value: Very High

Context: 68%

NBO Score:

82

Final Decision

#1 Family Meal

#2 Aquarium

#3 Annual Membership

What-If Simulation

The user should be able to change one condition.

Example:

Change journey:

Cart → Post-Visit

New result:

#1 Annual Membership

because membership becomes more contextually appropriate after the completed visit.

This is exactly why TICVAI needs a combined personalization and context architecture.

AI Requirements for Board 5

AI shall support:

- Customer propensity scoring
 - Customer intent prediction
 - Personalized product ranking
 - Next-Best-Offer decisioning
-

- Preference learning
- Recommendation fatigue detection
- Expected acceptance
- Expected incremental revenue
- Membership propensity
- Upgrade propensity
- Cross-sell propensity
- Customer-level timing optimization

AI should answer:

What is this customer most likely to want?

What should we recommend next?

What is the probability of acceptance?

What is the expected commercial value?

Is this recommendation appropriate now?

Should we suppress something this customer repeatedly rejects?

Next-Best-Offer Decision Architecture

The core runtime flow should be:

Customer / Session

-

Basket

-

Journey Context

↓

Candidate Generation

Boards 2 & 3

↓

Eligibility

Can customer purchase?

↓

Availability

Can TICVAI sell it?

↓

Journey Context

Is this the right moment?

↓

Customer Personalization

What does this customer prefer?

↓

Propensity Models

How likely is acceptance?

↓

Commercial Strategy

What objective are we optimizing?

↓

Guardrails

What are we permitted to recommend?

↓

Ranking

↓

Next-Best-Offer

This layered approach is preferable to giving an LLM the entire catalogue and simply asking:

“What should we sell?”

AI Layer Recommendation

For TICVAI, I recommend that the **LLM should not be the primary real-time recommendation model**.

The real-time NBO layer should primarily use:

- Recommendation models
- Ranking models
- Propensity models
- Rules
- Product affinity
- Behavioral models

An LLM can then assist with:

- Explaining recommendations
- Generating configuration suggestions
- Natural-language analytics
- Commercial strategy assistance
- Creating human-readable recommendation reasons

This will make the architecture more predictable, faster, cheaper, and easier to govern.

Cold-Start Strategy

TICVAI must also handle new clients with little historical data.

Phase 1 — Rules

Use manually configured product relationships.

Phase 2 — Aggregate Behavior

Use:

- Product popularity
- Basket patterns
- Segment performance

Phase 3 — Personalization

Once sufficient data exists:

- Customer propensity
- Behavioral ranking
- Individual NBO

Phase 4 — Continuous Optimization

Models improve using recommendation outcomes.

This is particularly important for onboarding new TICVAI tenants.

Feedback Loop

Every recommendation should feed learning events back into the system:

Presented

↓

Viewed

↓

Selected

↓

Added

↓

Purchased

↓

Redeemed

or:

Declined / Ignored

This allows models to learn from actual customer behavior.

Important Privacy Principle

Personalization must use only customer data that TICVAI is permitted to use for the relevant purpose.

The engine should therefore consume:

- Consent status
- Preference status
- Permitted CRM attributes
- Applicable tenant privacy configuration

If personalization is not permitted or insufficient data exists:

Fallback

Use:

Contextual / Product-Affinity Recommendations

rather than personalized customer-level targeting.

Key Integrations

Board 1

Strategy and ranking objectives.

Board 2

Upsell candidates.

Board 3

Cross-sell candidates.

Board 4

Journey context.

CRM

Customer profile and behavior.

Membership & Loyalty

Tier, benefits and history.

Ticketing

Purchases and visits.

Promotions

Eligible offers.

Pricing

Authoritative price.

Capacity / Inventory

Sellability.

F&B / Retail

Product information.

Access Control

Visit/redemption context.

Analytics

Historical behavior.

AI Platform

Models and inference.

Roles & Permissions

Recommended roles:

- CRM Manager

- Commercial Manager
- Marketing Manager
- Revenue Manager
- E-Commerce Manager
- Data Scientist / AI Administrator
- Data Analyst
- Privacy / Compliance Role
- System Administrator
- Auditor

Sensitive Permissions

Separately govern:

- Enable customer-level personalization
 - Configure signals
 - Change model
 - Change confidence threshold
 - Activate AI ranking
 - Override suppression
 - Access customer-level explanation
 - Enable new personalization use case
-

Scope Boundary with Board 6

Board 5 decides:

What personalized recommendation should this customer receive?

Board 6 will determine:

Did the recommendation actually work, how do we prove it, which strategy/model wins, and how should the recommendation system be optimized?

So Board 6 will focus on:

- Experimentation
- A/B testing
- Attribution
- Performance analytics
- Model monitoring
- Governance
- Recommendation optimization

without rebuilding the NBO decision engine.

Board 5 Acceptance Criteria

Board 5 shall be considered complete when:

1. TICVAI provides a Personalization & NBO Command Center.
2. Customer recommendation context can be viewed subject to permissions.
3. Anonymous and known-customer recommendation modes are supported.
4. Customer transaction history can inform recommendations.
5. Visit history can inform recommendations.
6. Membership and loyalty can inform recommendations.
7. Recommendation response history can inform recommendations.
8. Current basket/context can inform recommendations.
9. Approved AI signals can be configured.
10. Signal data source and freshness can be displayed.
11. Propensity models are supported.
12. Upgrade propensity is supported.
13. Membership propensity is supported.

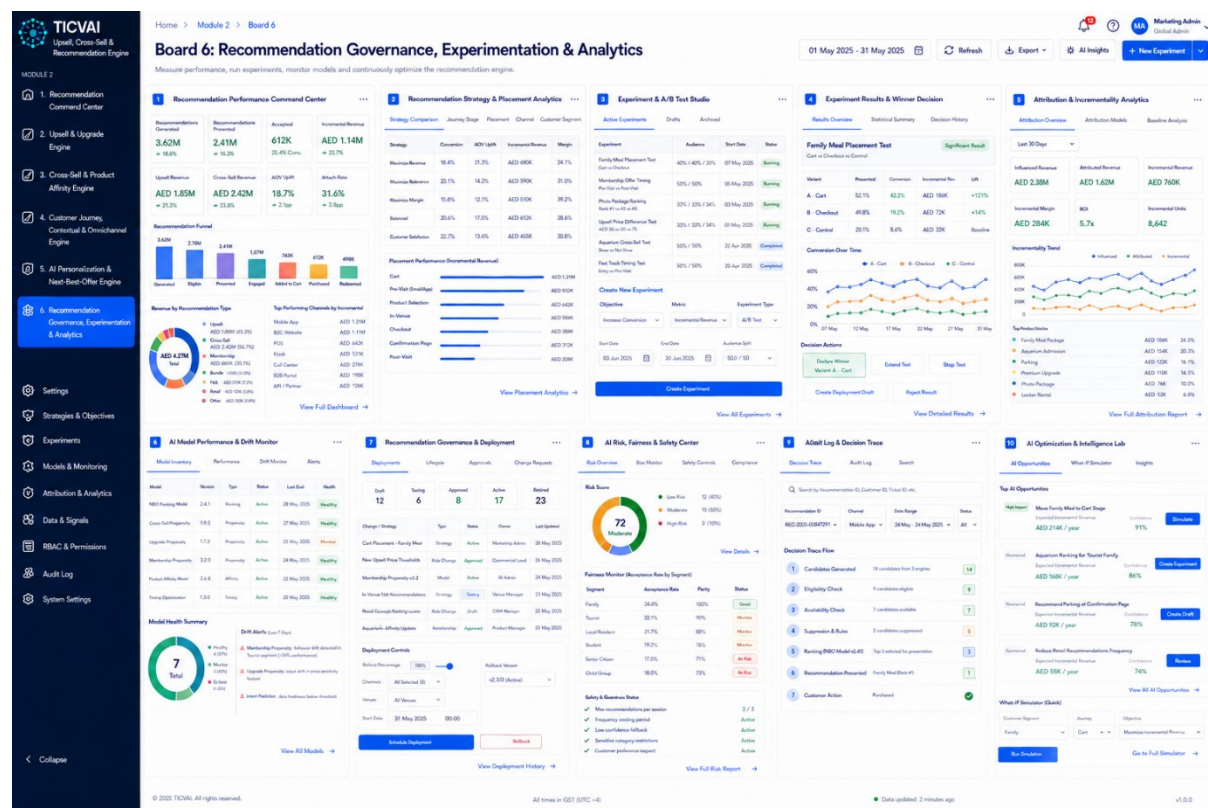
14. Cross-sell propensity is supported.
15. Multiple candidate sources can feed NBO.
16. Candidates are filtered through authoritative eligibility.
17. Candidates are filtered through availability.
18. Journey context affects ranking.
19. Customer-level personalization affects ranking.
20. Commercial objectives affect ranking.
21. Weighted decision policies are configurable.
22. Customer-specific recommendation fatigue can be detected.
23. Repeated declines can suppress recommendations.
24. Explicit customer preferences override inferred preferences where applicable.
25. Anonymous sessions receive contextual recommendations.
26. Known customers can receive deeper personalization.
27. Identity transition can preserve relevant journey state where permitted.
28. Every AI recommendation provides an explanation.
29. AI confidence is visible.
30. Model version is traceable.
31. Low-confidence AI can fall back to rule-based recommendation.
32. Models can be suspended without disabling the entire Recommendation Engine.
33. Administrators can simulate personalized customer scenarios.
34. What-if simulation is supported.
35. Cold-start tenants can operate without trained personalized models.
36. Recommendation feedback events support future model learning.
37. AI cannot override pricing, eligibility, inventory, capacity, consent or commercial governance.

- 38. Customer-level data access is controlled through TICVAI RBAC/PBAC.
- 39. Personalization respects applicable consent/preferences and tenant privacy configuration.
- 40. The NBO architecture can support future models without redesigning the customer-facing channels.

Board 5 is locked as: *AI Personalization & Next-Best-Offer Engine — 10 backend screens.*

Module 2 — Upsell, Cross-Sell & Recommendation Engine

Board 6 — Recommendation Governance, Experimentation & Analytics



Board Objective

Board 6 is the **final board of Module 2** and closes the recommendation lifecycle.

Boards 1–5 determine:

- **Board 1:** recommendation strategy and objectives.
- **Board 2:** upsell and upgrade opportunities.
- **Board 3:** cross-sell and product-affinity opportunities.
- **Board 4:** when and where recommendations should appear.
- **Board 5:** which recommendation is best for the specific customer.

Board 6 answers:

“Did the recommendation actually work, did it create incremental value, which strategy/model performs best, can we safely deploy changes, and how do we continuously improve the engine?”

It therefore provides the centralized **governance, experimentation, attribution, model monitoring, performance analytics, audit, and optimization layer** for the entire Recommendation Engine.

Board 6 shall contain 10 backend screens.

Screen 1 — Recommendation Performance Command Center

Purpose

Provide executive and operational visibility across the complete Recommendation Engine.

KPI Cards

- Recommendations Generated
- Recommendations Presented
- Recommendations Accepted
- Recommendation Conversion
- Upsell Revenue
- Cross-Sell Revenue
- Estimated Incremental Revenue
- AOV Uplift
- Attach Rate
- Incremental Margin
- AI Recommendation Share
- Rule-Based Recommendation Share
- Suppression Rate
- Active Experiments

- Model Health

Performance Funnel

Generated

↓

Eligible

↓

Presented

↓

Engaged

↓

Added to Cart

↓

Purchased

↓

Redeemed

Breakdown

Analyze by:

- Upsell
- Cross-sell
- Membership
- Bundle
- F&B
- Retail
- Experience
- Ancillary
- Channel

- Venue
- Customer segment

AI Executive Summary

Example:

Personalized recommendations generated AED 3.8M in influenced revenue this month, with an estimated AED 1.14M incremental contribution. Mobile App cross-sell recommendations produced the strongest incremental margin, while Checkout recommendations showed declining engagement.

Screen 2 — Recommendation Strategy & Placement Analytics

Purpose

Determine which recommendation strategies, placements, channels and journey stages perform best.

Strategy Comparison

Strategy	Conversion	AOV Uplift	Incremental Rev.	Margin
Max Revenue	18.4%	+21%	AED 680K	24%
Max Relevance	23.1%	+14%	AED 590K	31%
Max Margin	15.8%	+12%	AED 510K	39%

Placement Analytics

Compare:

- Product Page
- Ticket Selection
- Cart
- Checkout
- Confirmation
- Pre-Visit

- Mobile App
- In-Venue
- Post-Visit

Example Insight

Family Meal

Cart conversion:

42%

Checkout conversion:

19%

Recommendation:

Move Family Meal priority from Checkout to Cart for family segments.

Channel Analysis

Compare:

- B2C
 - App
 - POS
 - Kiosk
 - B2B
 - Call Center
 - API/Partner
-

Screen 3 — Recommendation Experiment & A/B Test Studio

Purpose

Allow TICVAI to scientifically test recommendation strategies rather than relying only on assumptions.

Experiment Types

Test:

- Recommendation product
- Recommendation sequence
- Ranking strategy
- AI vs Rules
- Journey placement
- Recommendation count
- Timing
- Customer segment
- Message/presentation
- Upsell threshold

Example Experiment

Objective

Increase Family Meal cross-sell.

Variant A

Recommend in Cart.

Variant B

Recommend during Checkout.

Control

No recommendation.

Audience Split

A — 40%

B — 40%

Control — 20%

Metrics

Measure:

- Conversion
 - Attach rate
 - Revenue
 - Incremental revenue
 - AOV
 - Margin
 - Customer response
 - Recommendation fatigue
-

Screen 4 — Experiment Results & Winner Decision Workspace

Purpose

Evaluate experiments and decide whether a recommendation strategy should be deployed.

Results Example

Variant	Conversion	Revenue	Incremental Rev.	Margin
Control	0%	—	—	—
Cart	42%	AED 440K	AED 186K	34%
Checkout	19%	AED 210K	AED 72K	35%

Statistical Information

Where sufficient data exists, show:

- Sample size
- Confidence
- Statistical significance
- Experiment duration

- Exposure balance

Winner

Variant A — Cart

Actions

Authorized users can:

- Declare Winner
- Extend Test
- Stop Test
- Reject Result
- Create Deployment Draft
- Send for Approval

AI Recommendation

Cart placement is the recommended winner. It generated +121% higher conversion with no material negative margin impact.

AI recommendation should not itself deploy the winner.

Screen 5 — Recommendation Attribution & Incrementality Analytics

Purpose

Determine whether a recommendation genuinely created additional commercial value.

This is essential.

A customer purchasing a recommended product does **not automatically mean the recommendation caused the purchase.**

Attribution Methods

Support, where data permits:

- Direct interaction attribution

- Recommendation tracking ID
- Last recommendation
- First recommendation
- Control-group comparison
- A/B experiment
- Pre/post comparison
- Matched audience
- AI-estimated incrementality

Example

Recommendation-influenced revenue:

AED 2.4M

Estimated baseline:

AED 1.65M

Estimated incremental revenue:

AED 750K

Metrics

- Influenced Revenue
- Attributed Revenue
- Incremental Revenue
- Incremental Units
- Incremental Margin
- Recommendation Cost
- ROI

Attribution Window

Configure:

- Same session

- 24 hours
- 7 days
- Until visit
- Custom

Different recommendation types may use different attribution windows.

Screen 6 — AI Model Performance & Drift Monitor

Purpose

Monitor whether recommendation and propensity models continue performing correctly after deployment.

This screen is critical if TICVAI wants a serious AI architecture.

Model Inventory

Show:

- Model
- Version
- Purpose
- Deployment date
- Status
- Last evaluation
- Data freshness

Models Could Include

- NBO Ranking
- Cross-Sell Propensity
- Upgrade Propensity
- Membership Propensity
- Product Affinity

- Recommendation Timing
- Customer Response

Performance Metrics

Depending on model type:

- Prediction accuracy
- Precision
- Recall
- Ranking quality
- Acceptance calibration
- Confidence
- Coverage

Drift Monitoring

Detect:

- Input drift
- Behavioral drift
- Performance drift
- Segment drift
- Product mix change

Example

Membership Propensity Model v3.2 has experienced a 14% reduction in predictive performance for Tourist segments during the last six weeks.

Recommended Action

Retrain / Review Model

Screen 7 — Recommendation Governance & Deployment Control

Purpose

Control how new strategies, rules, models and AI changes move into production.

Governance Lifecycle

Draft

↓

Testing

↓

Simulation

↓

Approval

↓

Scheduled

↓

Active

↓

Monitor

↓

Retired

Governed Objects

Apply lifecycle to:

- Recommendation strategy
- Product relationship
- Ranking policy
- Propensity model
- AI model
- Journey strategy

- Experiment winner
- Suppression policy

Approval Rules

Example:

Minor ranking-weight change

→ Commercial Manager.

New AI model

→ AI Administrator + Commercial.

Customer-level signal change

→ Data/Privacy approval where required.

Deployment

Support:

- Immediate
- Scheduled
- Phased rollout
- Percentage rollout
- Channel rollout
- Venue rollout

Rollback

Every deployment should retain a previous stable version.

Rollback to Previous Version

must be available to authorized users.

Screen 8 — AI Risk, Fairness, Explainability & Safety Center

Purpose

Provide governance over how AI recommendation decisions behave across customer populations and business contexts.

Monitoring Areas

- Recommendation bias
- Segment performance differences
- Excessive recommendation frequency
- Low-confidence decisions
- Unexplained recommendation changes
- Inappropriate product relationships
- Customer preference violations
- Commercial guardrail violations

Explainability

For sampled decisions, show:

Recommended Product

Family Meal

Why?

1. Product Affinity
2. Family Basket
3. Historical Acceptance
4. Journey Stage
5. Availability

Confidence

89% — **High**

Comparison

Administrators should be able to compare performance across permitted operational segments to identify materially different model behavior requiring investigation.

Safety Controls

Allow:

- Disable model
 - Disable recommendation type
 - Disable specific relationship
 - Raise confidence threshold
 - Switch to rule-based fallback
 - Emergency suspend
-

Screen 9 — Recommendation Audit, Decision Trace & Investigation

Purpose

Provide transaction-level explainability for Customer Service, Commercial, Audit, and technical investigation.

Search

Search by:

- Recommendation ID
- Transaction
- Customer/session where permitted
- Ticket
- Product
- Date
- Channel
- Model
- Strategy

Recommendation Trace Example

REC-NBO-847291

Customer Context

↓

Candidates Generated: **14**

↓

Eligibility Passed: **9**

↓

Availability Passed: **7**

↓

Suppression Removed: **2**

↓

Ranked Candidates: **5**

↓

Final Presented: **3**

Final Ranking

1. Family Meal
2. Aquarium
3. Photo

Decision Details

For each:

- Source engine
- Rule
- Affinity
- Propensity
- AI score
- Strategy
- Model/version

- Confidence
- Price received
- Promotion received
- Availability result

Outcome

Family Meal:

Purchased

Aquarium:

Ignored

Photo:

Declined

This provides end-to-end explainability.

Screen 10 — AI Optimization & Recommendation Intelligence Lab

Purpose

Turn all Recommendation Engine data into actionable improvement opportunities.

This is the final optimization screen for Module 2.

AI Opportunity Categories

Product Relationships

Add Water Park → Family Meal.

Ranking

Increase Aquarium ranking for Tourist Family segment.

Journey

Move Parking recommendation to Confirmation.

Timing

Present Fast Track 24 hours before visit.

Suppression

Reduce Retail recommendation frequency.

Strategy

Use Margin Optimization during peak season.

Model

Retrain Membership Propensity model.

Recommendation Card

Example:

AI Optimization Opportunity

Current:

Family Meal shown at Checkout.

Proposed:

Move to Cart.

Evidence

Current Conversion:

19.2%

Predicted Conversion:

37.8%

Expected Incremental Revenue:

+AED 214K/year

Margin Impact:

+AED 71K

Confidence:

91%

Actions

- Simulate
 - Create Experiment
 - Create Draft
 - Send for Approval
 - Dismiss
 - Snooze
-

Continuous Optimization Loop

Board 6 should complete the following lifecycle:

Recommend

↓

Present

↓

Track

↓

Purchase / Decline / Ignore

↓

Attribute

↓

Measure

↓

Experiment

↓

Learn

↓

AI Optimize

↓

Govern

↓

Deploy

↓

Recommend Again

This is what turns TICVAI from a static rule-based recommendation platform into a **continuously improving commercial decision engine**.

Critical Distinction: Revenue vs Incremental Revenue

It should implement this distinction carefully.

Influenced Revenue

Revenue from transactions involving a recommendation.

Attributed Revenue

Revenue assigned to a recommendation according to a defined attribution model.

Incremental Revenue

Estimated additional revenue that would **not have occurred without the recommendation**.

Example:

Recommendation-influenced sales:

AED 5M

does not mean:

Recommendation created AED 5M.

If expected baseline purchases were AED 4M:

Estimated incremental contribution may be:

AED 1M.

Board 6 must make this distinction visible.

Recommendation Tracking Architecture

Every recommendation should carry a persistent tracking identifier.

Example:

REC-2026-00847291

The tracking record should connect:

Customer/Session Context

↓

Candidate Set

↓

Recommendation Decision

↓

Channel / Placement

↓

Impression

↓

Interaction

↓

Basket

↓

Transaction

↓

Redemption

This enables accurate attribution and investigation.

Experiment Architecture

I recommend TICVAI treat experiments as first-class governed objects.

An experiment should contain:

- Experiment ID
- Hypothesis
- Strategy
- Population
- Variants
- Control
- Allocation
- Primary KPI
- Secondary KPI
- Guardrails
- Start/end date
- Statistical criteria
- Owner
- Approval
- Result
- Winner
- Deployment decision

This will be significantly stronger than simply adding an "A/B Test" checkbox to recommendation rules.

AI Model Governance

Every production model should have:

- Model ID

- Model type
- Version
- Owner
- Training dataset reference
- Training period
- Validation result
- Deployment date
- Confidence thresholds
- Applicable tenants/use cases
- Status
- Rollback version
- Performance history

Model Lifecycle

Development

→ **Validation**

→ **Approved**

→ **Production**

→ **Monitor**

→ **Retrain**

→ **Retire**

Human-in-the-Loop Principle

I recommend TICVAI separate AI actions into three levels:

Level 1 — AI Insight

AI identifies an opportunity.

No system change.

Level 2 — AI Draft

AI can create:

- Draft strategy
- Draft relationship
- Draft experiment
- Draft ranking change

Human approval is required.

Level 3 — Governed Optimization

For low-risk parameters, AI may automatically optimize within **pre-approved boundaries**.

Example:

Approved ranking range:

20%–30%

AI can adjust:

25% → 27%.

But AI cannot independently:

- Add a new sensitive data signal
 - Change a financial authorization
 - Override consent
 - Override eligibility
 - Remove mandatory guardrails
 - Deploy an unapproved model
-

Relationship with Promotions Analytics

There will naturally be some interaction between:

Promotions Board 10

and

Recommendation Board 6.

But they measure different things.

Promotions Analytics

Answers:

Did the **promotion/discount/bundle** perform?

Recommendation Analytics

Answers:

Did presenting **this product/offer to this customer at this moment** cause additional behavior?

Example:

Aquarium 15% Promotion

Promotion analytics measures the campaign.

Recommendation analytics measures whether recommending Aquarium in the Water Park cart increased Aquarium attachment.

Both systems can therefore analyze the same transaction from different perspectives without duplicating responsibility.

Relationship with Enterprise Reporting / Power BI

Board 6 provides operational recommendation analytics.

Enterprise BI should be able to consume its governed metrics.

Recommended outputs:

- Recommendation Revenue
- Upsell Revenue
- Cross-Sell Revenue
- Recommendation Conversion
- Attach Rate

- Incremental Revenue
- Incremental Margin
- NBO Acceptance
- Model Performance
- Experiment Performance

This enables management to combine Recommendation Engine metrics with:

- Ticketing
- F&B
- Retail
- CRM
- Membership
- Finance
- Promotions

in the enterprise reporting layer.

Natural-Language AI Analytics

Board 6 should support questions such as:

“Which three cross-sell recommendations generated the highest incremental margin last month?”

“Why did membership recommendation conversion decrease?”

“Compare Mobile App and B2C upsell performance.”

“Which recommendation strategy performs best for families?”

“Show recommendations with high impressions but low conversion.”

The AI should return:

Answer + KPI + filters + period + supporting analytical context

rather than an unsupported narrative.

Key Integrations

Board 1

Strategy and objectives.

Board 2

Upsell recommendations.

Board 3

Cross-sell relationships.

Board 4

Journey and context.

Board 5

NBO and personalization.

CRM

Customer and segment analysis.

Ticketing

Transactions and products.

Membership & Loyalty

Conversion behavior.

Promotions

Offer performance.

F&B / Retail

Cross-category transactions.

Pricing

Commercial values.

Finance

Margin and profitability.

Access Control

Redemption outcomes.

Enterprise BI

Reporting.

AI Platform

Models, inference and monitoring.

Roles & Permissions

Recommended roles:

- Executive
- Commercial Manager
- Revenue Manager
- Marketing Manager
- CRM Manager
- E-Commerce Manager
- Data Analyst
- Data Scientist / AI Administrator
- Privacy / Compliance
- Auditor
- System Administrator

Sensitive Permissions

Separately control:

- Create experiment
- Stop experiment
- Declare experiment winner
- Deploy recommendation change

- Activate AI model
 - Suspend AI model
 - Change confidence threshold
 - Enable governed auto-optimization
 - Roll back deployment
 - Access customer-level decision trace
-

Module 2 Architecture After Board 6

The completed module now becomes:

Board 1

Recommendation Command Center & Strategy

Defines the objective.

↓

Board 2

Upsell & Upgrade Recommendation Engine

Finds better/higher-value alternatives.

↓

Board 3

Cross-Sell & Product Affinity Engine

Finds complementary products.

↓

Board 4

Customer Journey, Contextual & Omnichannel Engine

Determines the right moment and channel.

↓

Board 5

AI Personalization & Next-Best-Offer Engine

Determines the best offer for the individual customer.

↓

Board 6

Governance, Experimentation & Analytics

Measures, learns, governs and improves the entire engine.

Acceptance Criteria

Board 6 shall be considered complete when:

1. Recommendation performance is visible centrally.
2. Upsell and cross-sell performance can be separated.
3. Recommendation funnels are measurable.
4. Recommendation performance can be analyzed by venue, channel, product and segment.
5. Journey-stage performance can be compared.
6. Placement performance can be compared.
7. Strategy performance can be compared.
8. Recommendation experiments can be created.
9. Control groups are supported.
10. Multiple experiment variants are supported.
11. Audience allocation is configurable.
12. Experiment KPIs and guardrails can be defined.
13. Experiment results show appropriate confidence/statistical information.
14. Experiment winners can be moved into governed deployment.
15. Influenced revenue can be measured.
16. Attributed revenue can be measured.

17. Estimated incremental revenue can be distinguished from influenced revenue.
18. Incremental margin can be measured where required cost data exists.
19. Attribution windows can be configured.
20. Recommendation tracking IDs support end-to-end attribution.
21. Production AI models can be inventoried.
22. Model versions are traceable.
23. Model performance can be monitored.
24. Model/data drift can be detected.
25. Models can be suspended.
26. Rule-based fallback can be activated.
27. Recommendation strategies and models support governed lifecycle states.
28. Phased deployments are supported.
29. Rollback to a previous approved version is supported.
30. AI decision explainability is available.
31. Recommendation confidence is visible where applicable.
32. Recommendation decision traces can be investigated.
33. Candidate rejection and suppression reasons are retained.
34. AI can identify optimization opportunities.
35. AI recommendations can create simulations, experiments or drafts rather than bypass governance.
36. Recommendation fatigue and customer-experience risks can be monitored.
37. Model/strategy changes are audited.
38. Customer-level traces are protected through RBAC/PBAC.
39. Enterprise BI can consume governed recommendation KPIs.

40. Natural-language analytics can query governed recommendation metrics with supporting context.

Board 6 is locked as: *Recommendation Governance, Experimentation & Analytics* — 10 backend screens.

With this board, **Module 2 — Upsell, Cross-Sell & Recommendation Engine is complete: 6 boards / 60 backend screens.**

The next planned module is **Module 3 — Payment & Payment Orchestration**, which we structured as **8 boards / approximately 80 backend screens.**